Mathematical Model and Data Model for Up-To-Four-Wire Distribution System OPF

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Abstract

Typical distribution systems have a nontrivial level of voltage and current unbalance due to single-phase loads, untransposed lines, and single-phase sections, and so physics-based optimal power flow (OPF) problems require the consideration of individual conductors. We outline the mathematical and data model developed by *IEEE Task Force on Benchmarking Multiconductor OPF for Distribution Systems* for use in development of benchmark multiconductor problems. The models are intended to be accessible and representative, covering common branch and nodal elements with appropriate constraints and objective functions to that define non-trivial OPF problems for distribution systems.

Document history

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  - Initial Mathematical and Data model shared with independent reviewers.
- V0.2; 13/4/2026
  - Incorporated comments from early reviewers and shared with wider Task Force membership.
- V0.2.1; 10/6/2026
  - Working draft starting to incorporate feedback from TF membership
- V0.2.2; 21/7/2026
  - First public launch of BMOPF math and data spec and JSON schema.

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1 Background

1.1 Motivation

The past three decades has seen a transformation in the availability and scalability of optimization solvers, expanding the problems that can be reliably solved from linear-programming methods to a wide range of non-linear and convex problems, covering new numerical methods and even computation paradigms (AI- and quantum-based solvers). At the same time, the number of problems of interest to utilities which can be formulated clearly as network-constrained optimization problems (a.k.a., optimal power flow (OPF) problems) has expanded considerably. Problems which have been solved using optimization approaches include analysis techniques such as (non-linear) power flow or distribution system state estimation; classical distribution system optimization problems such as optimal volt-var control; as well as future-looking problems such as peer-to-peer energy trading, distributed energy resource (DER) scheduling, Dynamic Operating Envelopes, or optimal droop settings for inverter-based resources (IBRs).

There is, however, little standardization in terms of the OPF-type problems for distribution networks that are presented and analyzed in the literature. In part, this is due to the relatively low numbers of freely available distribution network models, limited number of such datasets which have DERs integrated within those networks, and relatively low levels of controllable elements in legacy distribution networks. This leads to papers typically relying on modified test cases, rather than there being enough variation within the feeders to find an appropriate problem for new algorithms. This contrasts strongly with computational/numerical analysis fields, which typically develop and work with extensive benchmarking libraries.

This document is a community effort, developed by the 'Task Force on Benchmarking Multi-conductor OPF (BMOPF) for Distribution Systems', to provide a common mathematical model for distribution system OPF problems, alongside a library of reference problems in a data format convenient for research use, to enable direct comparisons between different approaches. All models and data are provided with appropriate licences to enable open sharing and modification by users.

Scope: Beyond Classical OPF

Whilst the term *optimal power flow* is used throughout this document, the ambition of the Task Force extends well beyond the classical OPF problem of minimising generation cost subject to network constraints. The unifying theme of the benchmark problems targeted here is the need to accurately represent distribution network physics rather than any particular objective function. The intent is that this serves as a reusable foundation across all such problem classes, not merely as infrastructure for a single OPF problem definition.

Generation cost minimisation, introduced as the objective function for this specification, is a convenient and well-posed starting point: it admits a unique solution, is straightforward to compare across solvers, and is equivalent to loss minimisation under mild conditions. Nevertheless, the same network physics underpins the aforementioned network-constrained optimisation problems relevant to distribution networks.

1.1.1 Previous Efforts Distribution System Analysis and OPF Benchmarks

Within the IEEE Power and Energy Society (PES) community, there have been efforts in developing benchmarks for transmission system OPFs, such as PGlib [1]. Only positive sequence cases are presented in PGLib, and they are limited to transmission applications. These are therefore not applicable to unbalanced distribution networks (the pitfalls of using positive sequence-only models for unbalanced networks are well known). The mathematical model presented builds on the earlier effort [2] (implemented as PowerModelsDistribution.jl tool [3]), extending the math model to capture the most common element configurations seen and carefully consider how elements can be connected to ground.

For several decades the IEEE Distribution Test Feeder Working Group has grown the number and types of networks included in its library of test feeders, culminating in the 8500 node test feeder released by the WG approximately a decade ago [4]. The purpose of those feeders was to provide relatively challenging test cases that enable researchers to test aspects such as high loading levels [4]. These models are well-known, but an open data license is not a requirement for Test Feeder cases, and today licences are not provided with those models, limiting permissible uses.

1.2 Features of the Models Provided

The Taskforce's philosophy is to make the cases reproducible, open, and accessible. The mathematical, data, and network models provided include the following features from the initial release.

- A permissive open data licence provided with all test cases.
- The use of the JSON file format, with a corresponding schema.
- SI units for physical quantities.
- A mix of construction types to capture network construction seen in various geographies.

From a technical perspective, the test cases will focus on low voltage (LV), medium voltage (MV) or high voltage (HV) distribution (due to higher levels of unbalance as compared to subtransmission). At present, we support:

- meshed networks,
- electrically parallel lines,
- 1-wire to 4-wire line models,
- perfect grounding and grounding through impedances.

To accommodate these, we use 'wire' coordinates so that the data format is flexible and extendable (the aspects supported would not be possible using sequence components only).

1.2.1 Present limitations

The aim of the TF is to provide a well-defined set of features that are universally required for distribution system OPF. Therefore, it will not be the intention to capture all possible types of equipment and optimization goals.

In this Version, we support a restricted set of features, with the following restrictions expected to be relaxed as priorities:

- Generator cost minimization (equivalent to network loss minimization) as the only objective function.
- Wye-delta, single-phase and centre tap ('split-phase') transformers types only.
- Static elements only (e.g., no controls for on-load tap changers or Inverter Based Resources (IBRs)).

1.2.2 Data Model Choices

JSON is chosen as it can easily by parsed from different programming languages. JSON is based on key-value pairs, and the keys are not ordered. Data extensions into an existing model can therefore be performed by adding nested entries. For compatibility across programming languages, we define quantities in real numbers, rather than complex numbers. Section 4 defines the Data Model fully.

The data model has common aspects with other data models used for distribution system analysis, but also differs in ways that may be unintuitive to users of other common tools. For example, the Data Model defines an explicit list of buses, enabling each to have their own voltage bounds and associated properties. This contrasts with OpenDSS, whereby buses are defined implicitly by the relations between elements. As compared to Matpower, we define *unique labels* (ids, as strings) for buses, generators, loads, etc, (i.e., we do not force (sequential) integers).

1.2.3 Out of Scope

The proposed format will not attempt to replace the Common Information Model (CIM). Instead the models aim to be a practical and lightweight approach for OPF problems for developers and researchers. Software to run the test cases will not be provided.

2 Terminology and Sets

The mathematical model and data model both use sets extensively to collect the elements of a network and the relationships between them. In this section, we define these sets and corresponding terminology, and present a basic network to illustrate these sets. Table 1 defines symbol formats (colors and typefaces) and other notation used throughout the document.

2.1 Sets and their Indices

Sets and the indices used to document elements of those sets are summarized in Table 2. Sets for buses, lines, transformers, switches, voltage sources, generators, loads, shunts, phases, and bus terminals are finite sets whose elements are indexed by a unique ID (represented in the data model as a string). Branch elements and nodal elements are defined in Table 3.

Symbol	Symbol type or meaning
x	real scalar variable
$\mathbf{X}$	real vector or matrix variable
x	complex scalar variable
$\mathbf{X}$	complex vector or matrix variable
x	real scalar parameter
$\mathbf{X}$	real vector or matrix parameter
x	complex scalar parameter
$\mathbf{X}$	complex vector or matrix parameter
x	string parameter
$\mathbf{X}$	array of string parameters
$\mathcal{X}$	set
$\mathbf{X}^{\mathrm{T}}$	transpose of $\mathbf{X}$
$\mathbf{X}^{*}$	conjugate of $\mathbf{X}$
$\mathbf{X}^{\mathrm{H}} = (\mathbf{X}^{*})^{\mathrm{T}}$	conjugate transpose of $\mathbf{X}$
$\circ$	element-wise multiplication
$\oslash$	element-wise division
j	imaginary unit, satisfies $j^2 = -1$
$a\angle b$	polar notation of complex number $a \cdot e^{jb}$
$\Re(x)$	real part of x
$\Im(x)$	imaginary part of x
$\mathbb{R}^{n\times m}$	set of real $n \times m$ matrices
$\mathbb{C}^{n\times m}$	set of complex $n \times m$ matrices
$\mathbf{1} \in \mathbb{R}^{n\times 1}$	vector of ones
$\mathbf{0} \in \mathbb{R}^{n\times 1}$	vector of zeros

Table 1: Typography and mathematical notation

Symbol	Represents	Primary index	Alternative index
$\mathcal{Z}^+$	Set of nonnegative integers	z	-
$\mathcal{P}$	Set of $p$hases, e.g., $\{a, b, c\}$	p	q
$\mathcal{N}$	Set of terminals, e.g., $\mathcal{P} \cup \{n\}$	p	q
$\mathcal{F}$	Set of nodal con$f$igurations, $\{\curlywedge, \Delta, \Phi\}$	f	-
$\mathcal{I}$	Set of buses	i	j
$\mathcal{L}$	Set of $l$ines	l	-
$\mathcal{X}$	Set of $t$ransformers	x	-
$\mathcal{W}$	Set of s$w$itches	w	-
$\mathcal{S}$	Set of voltage $s$ources	s	-
$\mathcal{G}$	Set of $g$enerators	g	-
$\mathcal{D}$	Set of loa$d$s ($d$emand)	d	-
$\mathcal{K}$	Set of capacitors	m	-
$\mathcal{H}$	Set of s$h$unts	h	-

Table 2: Sets and indices used to refer to their members

General classification	Relevant sets
Branch elements	Lines $\mathcal{L}$, Transformers $\mathcal{X}$, Switches $\mathcal{W}$
Nodal elements	Generators $\mathcal{G}$, Loads $\mathcal{D}$, Shunts $\mathcal{H}$, Voltage sources $\mathcal{S}$, Capacitors $\mathcal{K}$

Table 3: Classification of sets

2.1.1 Topology Sets, to link Branches to Buses

Topology sets are defined to capture the linking of buses through a branch element.

For indexing elements that together define the network topology, a triple index is used to allow for parallel lines, transformers and switches. Topologies are defined for basic (forward) and reverse senses. The forward line topology is,

$$lij \in \mathcal{T}^{L\rightarrow} \subset \mathcal{L} \times \mathcal{I} \times \mathcal{I}. \tag{1}$$

The reverse topology for lines can be defined as derived from the forward topology as

$$\mathcal{T}^{\mathrm{L}\leftarrow} = \{lji \mid lij \in \mathcal{T}^{L\rightarrow}\}. \tag{2}$$

Similarly, basic and reverse topologies for switches are,

$$wij \in \mathcal{T}^{W\rightarrow} \subset \mathcal{W} \times \mathcal{I} \times \mathcal{I}\,, \tag{3}$$
$$\mathcal{T}^{\mathrm{W}\leftarrow} = \{wji \mid wij \in \mathcal{T}^{W\rightarrow}\}\,, \tag{4}$$

and for transformers the topology is,

$$xij \in \mathcal{T}^{X\rightarrow} \subset \mathcal{X} \times \mathcal{I} \times \mathcal{I}\,, \tag{5}$$
$$\mathcal{T}^{\mathrm{X}\leftarrow} = \{xji \mid xij \in \mathcal{T}^{X\rightarrow}\}. \tag{6}$$

Assuming unique ids for each index x, w, l, the basic (forward) and reverse network topologies are found as the union of the topologies from the power conducting elements, i.e.,

$$\mathcal{T}^{\rightarrow} = \mathcal{T}^{L\rightarrow} \cup \mathcal{T}^{W\rightarrow} \cup \mathcal{T}^{X\rightarrow}, \tag{7}$$

$$\mathcal{T}^{\leftarrow} = \mathcal{T}^{\mathrm{L}\leftarrow} \cup \mathcal{T}^{\mathrm{W}\leftarrow} \cup \mathcal{T}^{\mathrm{X}\leftarrow}. \tag{8}$$

Finally, combined topologies are found as the union of basic and reverse topologies,

$$\mathcal{T}^{L} = \mathcal{T}^{L\rightarrow} \cup \mathcal{T}^{\mathrm{L}\leftarrow}, \tag{9}$$

$$\mathcal{T}^{X} = \mathcal{T}^{X\rightarrow} \cup \mathcal{T}^{\mathrm{X}\leftarrow}, \tag{10}$$

$$\mathcal{T}^{W} = \mathcal{T}^{W\rightarrow} \cup \mathcal{T}^{\mathrm{W}\leftarrow}, \tag{11}$$

$$\mathcal{T} = \mathcal{T}^{\rightarrow} \cup \mathcal{T}^{\leftarrow}. \tag{12}$$

2.1.2 Connectivity, to link Nodal Elements to Buses

Connectivity defines how nodal elements are linked to buses. These are therefore indexed according to an element index and a bus index. For voltage sources, loads, generators, and shunts these are respectively therefore

$$si \in \mathcal{C}^{S} \subset \mathcal{S} \times \mathcal{I}, \tag{13}$$

$$di \in \mathcal{C}^{D} \subset \mathcal{D} \times \mathcal{I}, \tag{14}$$

$$gi \in \mathcal{C}^{G} \subset \mathcal{G} \times \mathcal{I}, \tag{15}$$

$$hi \in \mathcal{C}^{H} \subset \mathcal{H} \times \mathcal{I}. \tag{16}$$

Note that for this version of the mathematical model, only a single voltage source is permitted. The source bus set therefore has only one element,

$$i \in \mathcal{I}^{\text{source}} \subset \mathcal{I} \text{ with } |\mathcal{I}^{\text{source}}| = 1 \tag{17}$$

2.1.3 Terminal Mapping to map Element Terminals to a Bus's Nodes

A terminal mapping describes how the terminals of branch and nodal elements are mapped to the terminals of a bus. This ensures an explicit link between any non-uniform per-phase properties of those elements (e.g., the per-phase powers of a wye-connected load) and their effect on the power flow of the multiconductor network.

For nodal elements with a unique bus, the terminal mappings is captured as elements in a set indexed by that bus. For example, the load terminal mapping $\mathcal{M}^{D}$ is indexed as

$$dzp \in \mathcal{M}^{D} \subset \mathcal{D} \times \mathcal{Z}^{+} \times \mathcal{N}, \tag{18}$$

where the index z captures the order of the terminals on the load. Similar expressions hold for voltage source, generator and shunt terminal mappings.

For branch elements with two or more buses, a similar expression holds, except the set values are additionally indexed by the bus. For example, for lines, the terminal mappings are

$$lizp \in \mathcal{M}^{D} \subset \mathcal{L} \times \mathcal{I} \times \mathcal{Z}^{+} \times \mathcal{N}. \tag{19}$$

Similar expressions hold for switch and transformer terminal mappings.

Finally, there is a terminal mapping that describes how bus terminals are connected to ground, $\mathcal{M}^{\emptyset}$. As ground is defined as a common voltage reference across a network, this set is therefore only indexed by the bus and terminal id,

$$ip \in \mathcal{M}^{\emptyset} \subset \mathcal{I} \times \mathcal{N}\,. \tag{20}$$

Other elements in the math model which can have a connection to ground (lines and shunts) are considered to always have a ground connection. An implicit ground connection to those elements is therefore considered (so no set is defined to account for ground connections for lines and shunts).

2.1.4 Element Configuration

The internal topology of some elements follow some common configurations (or can be modeled as such). These configurations define the number of terminals of the element, and how the parameters of the element relate to currents and voltages at the element's terminals.

The most complicated configurations are for distribution transformers, which can have several multiphase windings, resulting in substantially different properties between corresponding transformer models. In this model version, we consider

- Single phase (with symbol Φ)
- Center-tap (a.k.a., 'split phase' transformer, with symbol Υ)
- Wye-delta (with symbol ($\curlywedge\Delta$))
- Delta-wye (with symbol ($\Delta\curlywedge$))

Due to the qualitative differences the models, each of these four models are treated as individual element types within the data model.

For nodal elements, the considered configurations are single-phase, three-phase wye (with midpoint return), and three-phase delta, represented using Φ, $\curlywedge$, Δ respectively. The supported configurations for the applicable element types are presented in Table 4, with those configurations indexed according to the element index. For example, for loads these are defined by the load index d and configuration index f as

$$df \in \mathcal{R}^{D} \subset \mathcal{D} \times \mathcal{F}\,. \tag{21}$$

A similar expression holds for generators.

Superscript for sets and variables denotes subsets of corresponding variables. For example, the set of loads $\mathcal{D}$ or transformers $\mathcal{X}$ is the union

$$\mathcal{D} = \mathcal{D}^{\Phi} \cup \mathcal{D}^{\curlywedge} \cup \mathcal{D}^{\Delta}\,, \qquad \mathcal{X} = \mathcal{X}^{\Phi} \cup \mathcal{X}^{(\curlywedge\Delta)} \cup \mathcal{X}^{\Upsilon}\,. \tag{22}$$

Table 5 collects the symbols used for topology, connectivity, terminal mappings, and element configuration.

Nodal configuration	Symbol	JSON format	Supported		
			Load	Generator*	Capacitor
Wye (with midpoint return)	$\curlywedge$	"WYE"	✓	✓	✓
Delta	Δ	"DELTA"	✓	×	✓
Single phase	Φ	"SINGLE_PHASE"	✓	×	✓

Table 4: Supported configurations across nodal element types. *Generator objects are defined as having variable active and/or reactive power injections; constant generation can be modelled as a negative Load (see Section 3.7).

Symbol	Represents	Member tuple
$\mathcal{T}^{L\rightarrow}$	Basic line topology	lij
$\mathcal{T}^{L\leftarrow}$	Reverse line topology	lji
$\mathcal{T}^{L}$	Combined line topology	lij
$\mathcal{T}^{X\rightarrow}$	Basic transformer topology	xij
$\mathcal{T}^{X\leftarrow}$	Reverse transformer topology	xji
$\mathcal{T}^{X}$	Combined transformer topology	xij
$\mathcal{T}^{W\rightarrow}$	Basic switch topology	wij
$\mathcal{T}^{W\leftarrow}$	Reverse switch topology	wji
$\mathcal{T}^{W}$	Combined switch topology	wij
$\mathcal{T}^{\rightarrow}$	Basic network topology	kij
$\mathcal{T}^{\leftarrow}$	Reverse network topology	kji
$\mathcal{T}$	Combined network topology	kij
$\mathcal{C}^{S}$	Voltage source-bus connectivity	si
$\mathcal{C}^{G}$	Generator-bus connectivity	gi
$\mathcal{C}^{D}$	Load-bus connectivity	di
$\mathcal{C}^{H}$	Shunt-bus connectivity	hi
$\mathcal{M}^{S}$	Voltage source terminal mappings	szp
$\mathcal{M}^{G}$	Generator terminal mappings	gzp
$\mathcal{M}^{D}$	Load terminal mappings	dzp
$\mathcal{M}^{H}$	Shunt terminal mappings	hzp
$\mathcal{M}^{L}$	Line terminal mappings	$lizp$
$\mathcal{M}^{T}$	Transformer terminal mappings	$xizp$
$\mathcal{M}^{W}$	Switch terminal mappings	$wizp$
$\mathcal{M}^{\emptyset}$	Ground-bus terminal mappings	ip
$\mathcal{R}^{D}$	Load configurations	df
$\mathcal{R}^{G}$	Generator configurations	gf

Table 5: Topology, connectivity, terminal mapping and element configuration sets

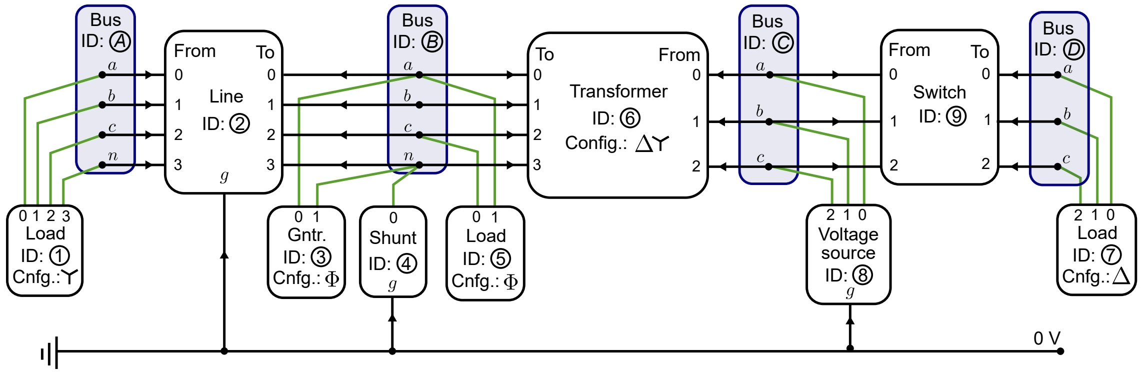

Figure 1: Example demonstrating how sets can be constructed from a network diagram.

2.2 A Network Example

Fig. 1 illustrates an example 4-bus network with single- and three-phase loads, a single phase generator, a neutral grounding shunt, a voltage source, connected together through a line, transformer, and switch, and a mixture of buses with and without grounded terminals.

The sets indices capturing branch and nodal elements are

$$\mathcal{I} = \{\textcircled{A}, \textcircled{B}, \textcircled{C}, \textcircled{D}\}\,, \tag{23}$$

$$\mathcal{L} = \{\textcircled{2}\}\,, \tag{24}$$

$$\mathcal{D} = \{\textcircled{1}, \textcircled{5}, \textcircled{7}\}\,, \tag{25}$$

$$\mathcal{G} = \{\textcircled{3}\}\,, \tag{26}$$

$$\mathcal{H} = \{\textcircled{4}\}\,, \tag{27}$$

$$\mathcal{X} = \{\textcircled{6}\}\,, \tag{28}$$

$$\mathcal{S} = \{\textcircled{8}\}\,, \tag{29}$$

$$\mathcal{W} = \{\textcircled{9}\}\,. \tag{30}$$

The forward topology for the lines, transformers, and switches are

$$\mathcal{T}^{L\rightarrow} = \{\textcircled{2}\textcircled{A}\textcircled{B}\}\,, \tag{31}$$

$$\mathcal{T}^{X\rightarrow} = \{\textcircled{6}\textcircled{C}\textcircled{B}\}\,, \tag{32}$$

$$\mathcal{T}^{W\rightarrow} = \{\textcircled{9}\textcircled{C}\textcircled{D}\}\,, \tag{33}$$

$$\tag{34}$$

noting how the orientation of this forward sense goes from the indicated 'From' to the indicated 'To'. The combined topology for the full network is

$$\mathcal{T} = \{\textcircled{2}\textcircled{A}\textcircled{B}, \textcircled{2}\textcircled{B}\textcircled{A}, \textcircled{6}\textcircled{B}\textcircled{C}, \textcircled{6}\textcircled{C}\textcircled{B}\}\,, \textcircled{9}\textcircled{C}\textcircled{D}, \textcircled{9}\textcircled{D}\textcircled{C}\}\,, \tag{35}$$

whilst each of the connectivity sets are

$$\begin{aligned}
\mathcal{C}^G &= \{\textcircled{3}\textcircled{B}\}\,, && (36)\\
\mathcal{C}^D &= \{\textcircled{1}\textcircled{A},\ \textcircled{5}\textcircled{B},\ \textcircled{7}\textcircled{D}\}\,, && (37)\\
\mathcal{C}^S &= \{\textcircled{8}\textcircled{C}\}\,, && (38)\\
\mathcal{C}^H &= \{\textcircled{4}\textcircled{B}\}\,. && (39)
\end{aligned}$$

The terminal mappings are

$$\begin{aligned}
\mathcal{M}^D &= \{\textcircled{1}0a,\ \textcircled{1}1b,\ \textcircled{1}2c,\ \textcircled{1}3n,\ \textcircled{5}0c,\ \textcircled{5}1a,\ \textcircled{7}0a,\ \textcircled{7}1b,\ \textcircled{7}2c\}\,, && (40)\\
\mathcal{M}^G &= \{\textcircled{3}0a,\ \textcircled{3}1n\}\,, && (41)\\
\mathcal{M}^H &= \{\textcircled{4}0n\}\,, && (42)\\
\mathcal{M}^S &= \{\textcircled{8}0a,\ \textcircled{8}1b,\ \textcircled{8}2c\}\,, && (43)\\
\mathcal{M}^L &= \{\textcircled{2}\textcircled{A}0a,\ \textcircled{2}\textcircled{A}1b,\ \textcircled{2}\textcircled{A}2c,\ \textcircled{2}\textcircled{A}3n,\ \textcircled{2}\textcircled{B}0a,\ \textcircled{2}\textcircled{B}1b,\ \textcircled{2}\textcircled{B}2c,\ \textcircled{2}\textcircled{B}3n\}\,, && (44)\\
\mathcal{M}^T &= \{\textcircled{6}\textcircled{B}0a,\ \textcircled{6}\textcircled{B}1b,\ \textcircled{6}\textcircled{B}2c,\ \textcircled{6}\textcircled{B}3n,\ \textcircled{6}\textcircled{C}0a,\ \textcircled{6}\textcircled{C}1b,\ \textcircled{6}\textcircled{C}2c\}\,, && (45)\\
\mathcal{M}^W &= \{\textcircled{9}\textcircled{C}0a,\ \textcircled{9}\textcircled{C}1b,\ \textcircled{9}\textcircled{C}2c,\ \textcircled{9}\textcircled{D}0a,\ \textcircled{9}\textcircled{D}1b,\ \textcircled{9}\textcircled{D}2c\}\,. && (46)
\end{aligned}$$

Finally, the load and generator configuration sets are

$$\begin{aligned}
\mathcal{R}^D &= \{\textcircled{1}\curlywedge,\ \textcircled{5}\Phi,\ \textcircled{7}\Delta\}\,, && (47)\\
\mathcal{R}^G &= \{\textcircled{3}\Phi\}\,. && (48)
\end{aligned}$$

3 Mathematical Model Specification

The mathematical model is based on a current-voltage ('IV') formulation. This has the advantage that the only nonlinearity introduced in the model is introduced at the constant power load and generator elements (or, if they are introduced, line or transformer thermal limits represented by an apparent power limit). Transformation matrices and constants used throughout this section are collected in Table 6.

The representation of the OPF mathematical model is non-unique. Where there are multiple equivalent potential representations of constraints (for example, constraints on the absolute value of a complex variable), a continuously differentiable representation is suggested. Further lifting of the equations to other variable spaces (e.g. power-voltage, or power-lifted-voltage as used on the conic relaxations) can be pursued but is not discussed.

3.1 Buses

Voltage-to-ground variables at bus i are,

$$\mathbf{U}_i = \begin{bmatrix} U_{i,a} \\ U_{i,b} \\ U_{i,c} \\ U_{i,n} \end{bmatrix}. \tag{49}$$

We can define voltages in polar or rectangular coordinates,

$$\mathbf{U}_i = \mathbf{U}_i^{\text{re}} + j\mathbf{U}_i^{\text{im}} = \mathbf{U}_i^{\text{mag}} \angle \mathbf{\Theta}_i, \tag{50}$$

where,

$$\mathbf{U}_i^{\text{re}} = \begin{bmatrix} U_{i,a}^{\text{re}} \\ U_{i,b}^{\text{re}} \\ U_{i,c}^{\text{re}} \\ U_{i,n}^{\text{re}} \end{bmatrix}, \mathbf{U}_i^{\text{im}} = \begin{bmatrix} U_{i,a}^{\text{im}} \\ U_{i,b}^{\text{im}} \\ U_{i,c}^{\text{im}} \\ U_{i,n}^{\text{im}} \end{bmatrix}, \mathbf{U}_i^{\text{mag}} = \begin{bmatrix} U_{i,a}^{\text{mag}} \\ U_{i,b}^{\text{mag}} \\ U_{i,c}^{\text{mag}} \\ U_{i,n}^{\text{mag}} \end{bmatrix}, \mathbf{\Theta}_i = \begin{bmatrix} \theta_{i,a} \\ \theta_{i,b} \\ \theta_{i,c} \\ \theta_{i,n} \end{bmatrix}. \tag{51}$$

The selection of the phase values is written,

$$\mathbf{U}_i[\mathcal{P}] = \begin{bmatrix} U_{i,a} \\ U_{i,b} \\ U_{i,c} \end{bmatrix}. \tag{52}$$

Symbol	Unit	Size	Represents	Value, or, definition
α	(-)	Scalar	Positive 120° unit rotation	$e^{2\pi/3}$
$\mathbf{F}$	(-)	3×3	Symmetrical component transformation matrix	(62)
$\mathbf{M}^{\perp}$	(-)	3×4	Phase-to-neutral transformation matrix	(55)
$\mathbf{M}^{\Delta}$	(-)	3×3	Phase-to-phase transformation matrix	(58)

Table 6: Constants

3.1.1 Voltage magnitude constraints

The bounds on voltage-to-ground magnitude are,

$$\mathbf{0} \leq \underbrace{\begin{bmatrix} U_{i,a}^{\min} \\ U_{i,b}^{\min} \\ U_{i,c}^{\min} \\ 0 \end{bmatrix}}_{\mathbf{U}_i^{\min}} \leq \begin{bmatrix} |U_{i,a}| \\ |U_{i,b}| \\ |U_{i,c}| \\ |U_{i,n}| \end{bmatrix} \leq \underbrace{\begin{bmatrix} U_{i,a}^{\max} \\ U_{i,b}^{\max} \\ U_{i,c}^{\max} \\ U_{i,n}^{\max} \end{bmatrix}}_{\mathbf{U}_i^{\max}}. \tag{53}$$

We can write this in a differentiable form as,

$$\mathbf{U}_i^{\min} \circ \mathbf{U}_i^{\min} \leq \mathbf{U}_i \circ \mathbf{U}_i{}^* \leq \mathbf{U}_i^{\max} \circ \mathbf{U}_i^{\max}. \tag{54}$$

where $\circ$ is the element-wise product and superscript $*$ indicates the conjugate (not conjugate transpose). Note that the lower bound is nonconvex (if nonzero), but the upper bound is convex.

The phase-to-neutral voltages are a linear combination of the phase-to-ground voltages,

$$\mathbf{U}_i^{\perp} = \begin{bmatrix} U_{i,an} \\ U_{i,bn} \\ U_{i,cn} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}}_{\mathbf{M}^{\perp}} \underbrace{\begin{bmatrix} U_{i,a} \\ U_{i,b} \\ U_{i,c} \\ U_{i,n} \end{bmatrix}}_{\mathbf{U}_i} = \mathbf{U}_i^{\perp,\text{mag}} \angle \mathbf{\Theta}_i^{\perp}. \tag{55}$$

The bounds on the phase-to-neutral voltage magnitudes are,

$$\underbrace{\begin{bmatrix} U_{i,an}^{\min} \\ U_{i,bn}^{\min} \\ U_{i,cn}^{\min} \end{bmatrix}}_{\mathbf{U}_i^{\perp,\min}} \leq \begin{bmatrix} |U_{i,an}| \\ |U_{i,bn}| \\ |U_{i,cn}| \end{bmatrix} \leq \underbrace{\begin{bmatrix} U_{i,an}^{\max} \\ U_{i,bn}^{\max} \\ U_{i,cn}^{\max} \end{bmatrix}}_{\mathbf{U}_i^{\perp,\max}}. \tag{56}$$

If the line is single-phase or triplex, the length is reduced accordingly. An equivalent smooth expression (quadratic) is,

$$\mathbf{U}_i^{\perp,\min} \circ \mathbf{U}_i^{\perp,\min} \leq \mathbf{U}_i^{\perp} \circ (\mathbf{U}_i^{\perp})^* \leq \mathbf{U}_i^{\perp,\max} \circ \mathbf{U}_i^{\perp,\max}. \tag{57}$$

The phase-to-phase voltages are,

$$\mathbf{U}_i^{\Delta} = \begin{bmatrix} U_{i,ab} \\ U_{i,bc} \\ U_{i,ca} \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ -1 & 0 & 1 & 0 \end{bmatrix} \underbrace{\begin{bmatrix} U_{i,a} \\ U_{i,b} \\ U_{i,c} \\ U_{i,n} \end{bmatrix}}_{\mathbf{U}_i} = \underbrace{\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix}}_{\mathbf{M}^{\Delta}} \underbrace{\begin{bmatrix} U_{i,a} \\ U_{i,b} \\ U_{i,c} \end{bmatrix}}_{\mathbf{U}_i[\mathcal{P}]}. \tag{58}$$

Bounds on phase-to-phase voltages are,

$$\underbrace{\begin{bmatrix} U_{i,ab}^{\min} \\ U_{i,bc}^{\min} \\ U_{i,ca}^{\min} \end{bmatrix}}_{\mathbf{U}_i^{\Delta,\min}} \leq \begin{bmatrix} |U_{i,ab}| \\ |U_{i,bc}| \\ |U_{i,ca}| \end{bmatrix} \leq \underbrace{\begin{bmatrix} U_{i,ab}^{\max} \\ U_{i,bc}^{\max} \\ U_{i,ca}^{\max} \end{bmatrix}}_{\mathbf{U}_i^{\Delta,\max}}. \tag{59}$$

The equivalent continuous expression (quadratic) is,

$$\mathbf{U}_i^{\Delta,\min} \circ \mathbf{U}_i^{\Delta,\min} \leq \mathbf{U}_i^{\Delta} \circ (\mathbf{U}_i^{\Delta})^* \leq \mathbf{U}_i^{\Delta,\max} \circ \mathbf{U}_i^{\Delta,\max}. \tag{60}$$

To simplify the definition of the transformation to symmetrical components, we first define a unit rotation α of positive 120 degrees,

$$\alpha = e^{j\frac{2\pi}{3}}. \tag{61}$$

Where three-phase loads are connected to a network, symmetrical components can sometimes be the most appropriate way to define voltage limits. For four-terminal (three-phase) buses, this is defined for the phase-to-neutral voltages,

$$\mathbf{U}_i^{012} = \begin{bmatrix} U_{i,0} \\ U_{i,1} \\ U_{i,2} \end{bmatrix} = \underbrace{\frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix}}_{\mathbf{F}} \begin{bmatrix} U_{i,an} \\ U_{i,bn} \\ U_{i,cn} \end{bmatrix} = \mathbf{F}\mathbf{U}_i^{\curlywedge}, \tag{62}$$

whilst for three-terminal (three-phase) buses, symmetrical components use phase-ground voltages

$$\mathbf{U}_i^{012} = \mathbf{F}\mathbf{U}_i. \tag{63}$$

We can now bound the magnitude of the symmetrical components,

$$\underbrace{\begin{bmatrix} 0 \\ U_{i,1}^{\min} \\ 0 \end{bmatrix}}_{\mathbf{U}_i^{012,\min}} \leq \begin{bmatrix} |U_{i,0}| \\ |U_{i,1}| \\ |U_{i,2}| \end{bmatrix} \leq \underbrace{\begin{bmatrix} U_{i,0}^{\max} \\ U_{i,1}^{\max} \\ U_{i,2}^{\max} \end{bmatrix}}_{\mathbf{U}_i^{012,\max}}. \tag{64}$$

Again, we can formulate a continuous (quadratic) expression,

$$\mathbf{U}_i^{012,\min} \circ \mathbf{U}_i^{012,\min} \leq \mathbf{U}_i^{012} \circ (\mathbf{U}_i^{012})^* \leq \mathbf{U}_i^{012,\max} \circ \mathbf{U}_i^{012,\max}. \tag{65}$$

3.1.2 Voltage angle constraints between buses

Bounds can be derived for bus pair angle differences [5]. For $ij \in \mathcal{B}$, these angle differences are

$$\boldsymbol{\Theta}_{ij} = \boldsymbol{\Theta}_i - \boldsymbol{\Theta}_j = \begin{bmatrix} \theta_{i,a} - \theta_{j,a} \\ \theta_{i,b} - \theta_{j,b} \\ \theta_{i,c} - \theta_{j,c} \\ \theta_{i,n} - \theta_{j,n} \end{bmatrix}. \tag{66}$$

Allowing us to define bounds on the angle differences too,

$$\begin{bmatrix} -\pi/2 \\ -\pi/2 \\ -\pi/2 \\ -\infty \end{bmatrix} \leq \underbrace{\begin{bmatrix} \theta_{ij,a}^{\min} \\ \theta_{ij,b}^{\min} \\ \theta_{ij,c}^{\min} \\ \theta_{ij,n}^{\min} \end{bmatrix}}_{\boldsymbol{\Theta}_{ij}^{\min}} \leq \underbrace{\begin{bmatrix} \theta_{i,a} - \theta_{j,a} \\ \theta_{i,b} - \theta_{j,b} \\ \theta_{i,c} - \theta_{j,c} \\ \theta_{i,n} - \theta_{j,n} \end{bmatrix}}_{\boldsymbol{\Theta}_{ij}} \leq \underbrace{\begin{bmatrix} \theta_{ij,a}^{\max} \\ \theta_{ij,b}^{\max} \\ \theta_{ij,c}^{\max} \\ \theta_{ij,n}^{\max} \end{bmatrix}}_{\boldsymbol{\Theta}_{ij}^{\max}} \leq \begin{bmatrix} \pi/2 \\ \pi/2 \\ \pi/2 \\ \infty \end{bmatrix}. \tag{67}$$

Angle difference bounds between phases on a single bus are,

$$\underbrace{\begin{bmatrix}\theta^{\min}_{i,abn}\\ \theta^{\min}_{i,bcn}\\ \theta^{\min}_{i,can}\end{bmatrix}}_{\boldsymbol{\Theta}^{\curlywedge,\min}_i} \leq \underbrace{\begin{bmatrix}\theta_{i,an}-\theta_{i,bn}-2\pi/3\\ \theta_{i,bn}-\theta_{i,cn}-2\pi/3\\ \theta_{i,cn}-\theta_{i,an}-2\pi/3\end{bmatrix}}_{\mathbf{M}^{\curlywedge}\boldsymbol{\Theta}_i-2\pi/3} \leq \underbrace{\begin{bmatrix}\theta^{\max}_{i,abn}\\ \theta^{\max}_{i,bcn}\\ \theta^{\max}_{i,can}\end{bmatrix}}_{\boldsymbol{\Theta}^{\curlywedge,\max}_i}. \tag{68}$$

3.1.3 Current Balance at a Bus

Kirchhoff's current law (KCL) at bus i is,

$$\underbrace{\sum_{lij\in\mathcal{T}^L}\mathbf{I}_{lij}}_{\text{lines}} + \underbrace{\sum_{xij\in\mathcal{T}^X}\mathbf{I}_{xij}}_{\text{transformers}} + \underbrace{\sum_{wij\in\mathcal{T}^W}\mathbf{I}_{wij}}_{\text{switches}} + \underbrace{\sum_{di\in\mathcal{C}^D}\mathbf{I}_{d}}_{\text{loads}} - \underbrace{\sum_{gi\in\mathcal{C}^G}\mathbf{I}_{g}}_{\text{generators}} + \underbrace{\sum_{hi\in\mathcal{C}^H}\mathbf{I}_{h}}_{\text{shunts}} + \underbrace{\sum_{mi\in\mathcal{C}^K}\mathbf{I}_{m}}_{\text{capacitors}} = \mathbf{0}. \tag{69}$$

For a four-terminal bus, the current vectors are, for example,

$$\mathbf{I}_{lij} \stackrel{\text{def}}{=} \begin{bmatrix}I_{lij,a}\\ I_{lij,b}\\ I_{lij,c}\\ I_{lij,n}\end{bmatrix}, \mathbf{I}_{xij} \stackrel{\text{def}}{=} \begin{bmatrix}I_{xij,a}\\ I_{xij,b}\\ I_{xij,c}\\ I_{xij,n}\end{bmatrix}, \mathbf{I}_{wij} \stackrel{\text{def}}{=} \begin{bmatrix}I_{wij,a}\\ I_{wij,b}\\ I_{wij,c}\\ I_{wij,n}\end{bmatrix}, \mathbf{I}_{d} \stackrel{\text{def}}{=} \begin{bmatrix}I_{d,a}\\ I_{d,b}\\ I_{d,c}\\ I_{d,n}\end{bmatrix}, \mathbf{I}_{g} \stackrel{\text{def}}{=} \begin{bmatrix}I_{g,a}\\ I_{g,b}\\ I_{g,c}\\ I_{g,n}\end{bmatrix}. \tag{70}$$

KCL (69) applies at all buses individually, except where a bus's terminals are connected to a voltage source (or a bus's terminal is grounded), in which case the corresponding terminal should not have KCL defined (or, an auxiliary current defined, as described in Section 3.9.1). Similarly, in case of a different set of bus terminals than $\{a, b, c, n\}$, then the dimensions of (70) should be updated accordingly.

3.1.4 Grounding at a Bus

Grounding at a bus refers to the connection of the terminals of a bus through a conductor to ground. The grounded terminals can *perfectly grounded* (i.e., connected to ground through a 0 impedance), *grounded through an impedance*, or *ungrounded*.

If all bus terminals are ungrounded, no further relationships or variables to be defined. A bus with one or more elements connected to ground through an impedance can be described using the model of a bus shunt (Section 3.5). For the buses with perfectly grounded terminals, the voltage at those terminals are constrained to have value zero,

$$\mathbf{U}_{i,p} = 0 \quad \forall\, ip \in \mathcal{M}^{\emptyset}\,. \tag{71}$$

At these bus terminals, KCL (69) should not be be applied (or, an auxiliary current introduced as described in Section 3.9.1).

3.2 Lines

Fig. 2 represents the Π-model of a four-wire power conductor, e.g. a cable or overhead line.

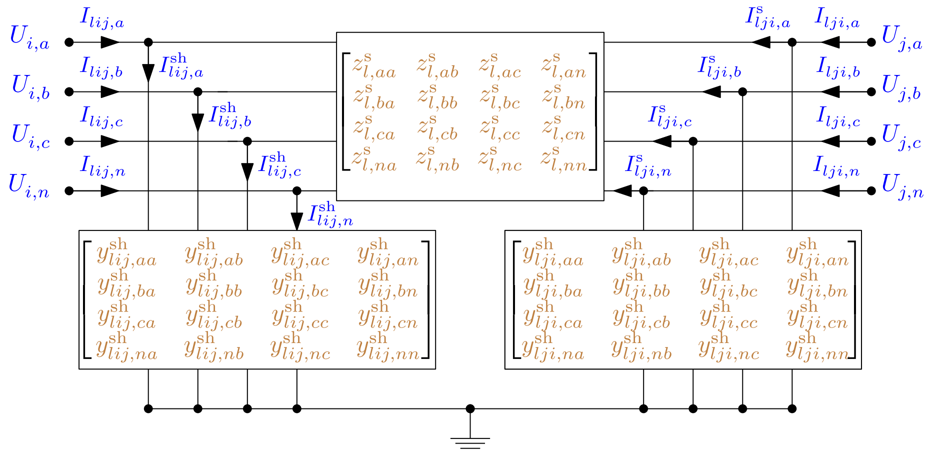

Figure 2: Four-wire line model with shunt admittance

The terminal currents flowing into a line l, at bus i in the direction of bus j, are

$$\mathbf{I}_{lij} = \begin{bmatrix} I_{lij,a} \\ I_{lij,b} \\ I_{lij,c} \\ I_{lij,n} \end{bmatrix}. \tag{72}$$

The current magnitude bounds are,

$$\mathbf{0} \leq \begin{bmatrix} |I_{lij,a}| \\ |I_{lij,b}| \\ |I_{lij,c}| \\ |I_{lij,n}| \end{bmatrix} \leq \underbrace{\begin{bmatrix} I^{\text{max}}_{lij,a} \\ I^{\text{max}}_{lij,b} \\ I^{\text{max}}_{lij,c} \\ I^{\text{max}}_{lij,n} \end{bmatrix}}_{\mathbf{I}^{\text{max}}_{lij}}. \tag{73}$$

A continuous and convex expression for the current bounds is,

$$\mathbf{I}_{lij} \circ (\mathbf{I}_{lij})^* \leq \mathbf{I}^{\text{max}}_{lij} \circ \mathbf{I}^{\text{max}}_{lij}. \tag{74}$$

The current flowing into the branch divides over the series ($\mathbf{I}^{\text{s}}_{lij}$) and shunt ($\mathbf{I}^{\text{sh}}_{lij}$) elements

$$\mathbf{I}_{lij} = \mathbf{I}^{\text{sh}}_{lij} + \mathbf{I}^{\text{s}}_{lij}. \tag{75}$$

We can substitute in the shunt admittance,

$$\mathbf{I}_{lij} = \mathbf{Y}^{\text{sh}}_{lij}\mathbf{U}_i + \mathbf{I}^{\text{s}}_{lij}. \tag{76}$$

At the receiving and we get a very similar expression,

$$\mathbf{I}_{lji} = \mathbf{Y}^{\text{sh}}_{lji}\mathbf{U}_j + \mathbf{I}^{\text{s}}_{lji}. \tag{77}$$

Conservation of current allows us to eliminate one of the directional series current variables through

$$\mathbf{I}^{\text{s}}_{lij} + \mathbf{I}^{\text{s}}_{lji} = 0\,. \tag{78}$$

The complex power flow into line l at bus i in the direction of bus j is,

$$\mathbf{S}_{lij} = \mathbf{U}_i \circ {\mathbf{I}_{lij}}^*\,. \tag{79}$$

We can define apparent power bounds as,

$$-\mathbf{S}^{\max}_{lij} \circ \mathbf{S}^{\max}_{lij} \leq \mathbf{S}_{lij} \circ {\mathbf{S}_{lij}}^* \leq \mathbf{S}^{\max}_{lij} \circ \mathbf{S}^{\max}_{lij}. \tag{80}$$

In current-voltage variables, this expression simplifies to,

$$-\mathbf{S}^{\max}_{lij} \circ \mathbf{S}^{\max}_{lij} \leq \mathbf{U}_i \circ (\mathbf{U}_i)^* \circ \mathbf{I}_{lij} \circ (\mathbf{I}_{lij})^* \leq \mathbf{S}^{\max}_{lij} \circ \mathbf{S}^{\max}_{lij}. \tag{81}$$

Ohm's law over a line l, between buses i and j is,

$$\forall lij \in \mathcal{T}^{L\rightarrow} : \mathbf{U}_j = \mathbf{U}_i - \mathbf{Z}^{\text{s}}_l \mathbf{I}^{\text{s}}_{lij}, \tag{82}$$

where $\mathbf{Z}^{\text{s}}_l$ is the series impedance (matrix) of line l.

The power flow through line l, in the direction of bus i to j is,

$$\mathbf{S}_{lij} = \mathbf{U}_i(\mathbf{U}_i)^*(\mathbf{Y}^{\text{sh}}_{lij})^* + \mathbf{U}_i(\mathbf{U}_i - \mathbf{U}_j)^*(\mathbf{Y}^{\text{s}}_l)^*\,. \tag{83}$$

3.3 LineCodes

A LineCode is an object which defines the per-unit-length impedance and admittance parameters for a conductor type with a given geometry, typically calculated using Carson's equations or a similar approach (see, e.g., [6]).

For a line l, we derive the line impedance from the linecode c, by multiplying the per-length impedance $\mathbf{Z}^{\text{s}}_c$ with the line length ℓ_l, and the shunt admittance $\mathbf{Y}^{\text{sh}}_c$ in proportion to the per-length admittance,

$$\mathbf{Z}^{\text{s}}_l \leftarrow \mathbf{Z}^{\text{s}}_c \cdot \ell_l, \quad \mathbf{Y}^{\text{sh}}_{lij} \leftarrow \mathbf{Y}^{\text{sh}}_c \cdot \ell_l/2, \quad \mathbf{Y}^{\text{sh}}_{lji} \leftarrow \mathbf{Y}^{\text{sh}}_c \cdot \ell_l/2. \tag{84}$$

Linecode current limits are copied to line current limits,

$$\mathbf{I}^{\max}_l \leftarrow \mathbf{I}^{\max}_c. \tag{85}$$

We allow for an optional `source` field in the data model. A free-text string recording the provenance of a linecode's impedance matrices — where the impedance values came from. It is descriptive metadata only: it has no effect on the electrical model, the power-flow/OPF equations, or validation, and may be omitted. Suggested values: "fem" (finite-element derived), "datasheet" (manufacturer data), "import" (carried over from a source model, e.g. OpenDSS). Tools may use it for reporting and audit trails; unrecognised values are permitted.

3.4 Loads

Fig. 3 illustrates the connection of single phase, wye-connected, and delta-connected loads. As compared to Generators (Section 3.7), Loads are considered to have fixed reference demand. Controllable loads (with variable active or reactive power injection) are defined using a Generator model.

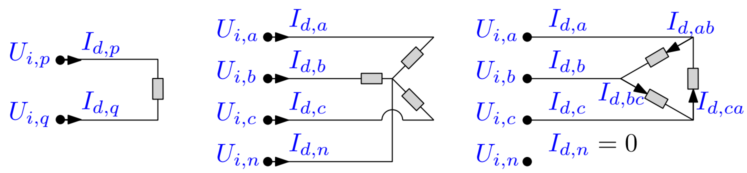

Figure 3: Variables for single phase (left), wye (center), and delta (right) loads.

3.4.1 Loads with Single-Phase Configuration

A single phase load d can be defined between any two nodes p, q of any single bus i. Conservation of current implies that

$$I_{d,p} + I_{d,q} = \mathbf{1}^{\mathrm{T}} \mathbf{I}_d = 0\,. \tag{86}$$

The complex power variable for the load $\mathbf{S}_d^{\Phi}$ is equal to its demand reference parameter for the load d, $\mathbf{S}_d^{\mathrm{ref},\Phi}$,

$$\mathbf{S}_d^{\Phi} = (\mathbf{U}_{i,p} - \mathbf{U}_{i,q}) {I_{d,p}}^{*} = \mathbf{S}_d^{\mathrm{ref},\Phi}\,. \tag{87}$$

3.4.2 Loads with Wye Configuration

Applying conservation of current to an wye load implies

$$I_{d,a} + I_{d,b} + I_{d,c} + I_{d,n} = \mathbf{1}^{\mathrm{T}} \mathbf{I}_d = 0\,. \tag{88}$$

For wye connected loads, we define the power variable,

$$\mathbf{S}_d^{\curlywedge} = \mathbf{U}_i^{\curlywedge} \circ (\mathbf{I}_d[\mathcal{P}])^{*}. \tag{89}$$

By convention, the complex power defined for a wye load $\mathbf{S}_d^{\mathrm{ref},\curlywedge}$ is equal to the complex power with neutral as reference, allowing us to set this equal to $\mathbf{S}_d^{\curlywedge}$,

$$\mathbf{S}_d^{\mathrm{ref},\curlywedge} = \mathbf{S}_d^{\curlywedge}\,. \tag{90}$$

3.4.3 Loads with Delta Configuration

From the perspective of the current being supplied externally, a delta load conserves current in the following fashion,

$$I_{d,a} + I_{d,b} + I_{d,c} = \mathbf{1}^{\mathrm{T}} \mathbf{I}_d[\mathcal{P}] = 0 \tag{91}$$

However, this current divides over the internal current flows of the delta load, $I_{d,ab}, I_{d,bc}, I_{d,ca}$, which allows for loop flows, therefore has an additional degree of freedom,

$$\underbrace{\begin{bmatrix} I_{d,a} \\ I_{d,b} \\ I_{d,c} \end{bmatrix}}_{\mathbf{I}_d[\mathcal{P}]} = \begin{bmatrix} I_{d,ab} - I_{d,ca} \\ I_{d,bc} - I_{d,ab} \\ I_{d,ca} - I_{d,bc} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}}_{(\mathbf{M}^{\Delta})^{\mathrm{T}}} \underbrace{\begin{bmatrix} I_{d,ab} \\ I_{d,bc} \\ I_{d,ca} \end{bmatrix}}_{\mathbf{I}_d^{\Delta}}, \tag{92}$$

due to rank($\mathbf{M}^{\Delta}$) = 2. Note that (92) implies (91), but (91) by itself is not enough to recover the correct values for $\mathbf{I}_d^{\Delta}$. Variables $\mathbf{I}_d[\mathcal{P}]$ are the values for the current that appear in KCL (69), though they are only linear combinations of $\mathbf{I}_d^{\Delta}$, so $\mathbf{I}_d[\mathcal{P}]$ could be eliminated.

Now we can define the (internal) load power of a delta load,

$$\mathbf{S}_d^{\Delta} = \mathbf{U}_i^{\Delta} \circ \mathbf{I}_d^{\Delta *}. \tag{93}$$

Finally, we assign a set point to these variables,

$$\mathbf{S}_d^{\text{ref},\Delta} = \mathbf{S}_d^{\Delta}\,. \tag{94}$$

3.4.4 Voltage Dependent Loads

Let $\Delta U_{d,k}$ be the voltage drop across sub-load k (phase-to-neutral for `WYE`, the terminal-pair difference for `SINGLE_PHASE`, line-to-line for `DELTA`). The nominal voltage of the load U_d^{nom} is defined as

- The line-to-line voltage for three-phase loads (wye- and delta-connected)
- The nominal voltage across the terminals of a single-phase load

Let $P_{d,k}^{\text{nom}}$, $Q_{d,k}^{\text{nom}}$ of a ZIPtype load be the nominal active and reactive powers respectively; then, the realised powers are

$$P_{d,k} = P_{d,k}^{\text{nom}} \left(\alpha_{d,k}^{Z} \frac{|\Delta U_{d,k}|^2}{(k_\beta U_d^{\text{nom}})^2} + \alpha_{d,k}^{I} \frac{|\Delta U_{d,k}|}{k_\beta U_d^{\text{nom}}} + \alpha_{d,k}^{P} \right), \tag{95}$$

$$Q_{d,k} = Q_{d,k}^{\text{nom}} \left(\beta_{d,k}^{Z} \frac{|\Delta U_{d,k}|^2}{(k_\beta U_d^{\text{nom}})^2} + \beta_{d,k}^{I} \frac{|\Delta U_{d,k}|}{k_\beta U_d^{\text{nom}}} + \beta_{d,k}^{P} \right). \tag{96}$$

where $k_\beta = 1$ for delta- and single-phase loads and $k_\beta = 1/\sqrt{3}$ for wye-connected loads (because the line-line voltage across a wye-connected load is split across the loads in wye configuration).

The classical single-mode models are the special cases $(\alpha_Z, \alpha_I, \alpha_P) = (1, 0, 0)$ (constant impedance, $P \propto |V|^2$), $(0, 1, 0)$ (constant current, $P \propto |V|$), and $(0, 0, 1)$ (constant power, recovering (90)).

The exponential model is

$$P_{d,k} = P_{d,k}^{\text{nom}} \left(\frac{|\Delta U_{d,k}|}{k_\beta U_d^{\text{nom}}} \right)^{\gamma_{d,k}^{P}}, \qquad Q_{d,k} = Q_{d,k}^{\text{nom}} \left(\frac{|\Delta U_{d,k}|}{k_\beta U_d^{\text{nom}}} \right)^{\gamma_{d,k}^{Q}}. \tag{97}$$

The values of k_β are as for the ZIP-type loads. Integer exponents $\gamma \in \{0, 1, 2\}$ coincide with the constant-power, constant-current and constant-impedance terms of (95)–(96) and are routed there to keep the formulation quadratic; only a genuinely non-integer exponent emits the nonlinear power term $P_{d,k}^{\text{nom}}\,(W_{d,k}/(U_d^{\text{nom}})^2)^{\gamma_{d,k}^{P}/2}$.

3.5 Bus shunt

A bus shunt refers to the connection of an admittance between terminals of a bus. Current flowing into a shunt h with admittance $\mathbf{Y}_h$ at bus i is,

$$\mathbf{I}_h = \mathbf{Y}_h \mathbf{U}_i. \tag{98}$$

Note that this model also permits shunts connected between terminals of a bus and ground.

The shunt admittance is a square matrix,

$$\mathbf{Y}_h = \begin{bmatrix} Y_{h,aa} & Y_{h,ab} & Y_{h,ac} & Y_{h,an} \\ Y_{h,ba} & Y_{h,bb} & Y_{h,bc} & Y_{h,bn} \\ Y_{h,ca} & Y_{h,cb} & Y_{h,cc} & Y_{h,cn} \\ Y_{h,na} & Y_{h,nb} & Y_{h,nc} & Y_{h,nn} \end{bmatrix} . \tag{99}$$

Note that there are no terms in (99) indexed in the ground phase g because the voltage at ground is zero. The matrix (99) can be singular or non-singular, but in all cases zero-filled rows and columns should be eliminated by adapting the terminals accordingly. The current flowing into ground at the shunt's bus is $\sum \mathbf{I}_h$.

We can use a shunt to ground the neutral through an impedance. In particular, a single-phase shunt,

$$\mathbf{Y}_h = \begin{bmatrix} Y_{h,nn} \end{bmatrix} , \qquad \mathbf{N}_h = \begin{bmatrix} n \end{bmatrix} , \tag{100}$$

connects an impedance $Z_{h,nn} = 1/Y_{h,nn}$ between neutral and ground terminals at a bus.

3.6 Capacitors

A capacitor models a fixed (nameplate-rated) shunt capacitor bank. It is a passive, constant-susceptance element with no decision variables. It is a special case of the bus shunt (Section 3.5) whose susceptance is derived from the nameplate rather than given as an explicit admittance matrix. The set of capacitors is $\mathcal{K}$, indexed by m.

For three-phase capacitors (with either wye or delta connection) the nameplate, nominal voltage U_m^{nom} refers to the line-to-line nominal voltage. For single phase capacitors, the nominal voltage U_m^{nom} refers to the nominal voltage across the terminals of the individual capacitor element.

Capacitors deliver the rated reactive power Q_m^{nom} when the voltage across its terminals match their nominal values. The susceptance of a capacitor is therefore

$$b_m = \frac{Q_m^{\text{nom}}}{\beta_{\text{cap}}(U_m^{\text{nom}})^2} , \tag{101}$$

where the constant β_{cap} takes value

- Single phase: $\beta_{\text{cap}} = 1$, as the nominal voltage is across the sole single-phase capacitor
- Delta-connected: $\beta_{\text{cap}} = 3$, as the reactive power generated is across three single-phase capacitors with nominal voltage across them
- Wye-connected: $\beta_{\text{cap}} = 1$, as the reactive power is generated from three single-phase capacitors which have voltage $\sqrt{3}$ less than nominal voltage

The elementary capacitors are assembled into a terminal-space susceptance matrix $\mathbf{B}_m$ by the reciprocal two-node stamp $b_m \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ applied across each connected terminal pair:

- Wye connected — one stamp per phase, between that phase and the neutral;
- Single phase — one stamp between the two terminals;

- Delta connected — one stamp per phase pair $(k,\ k \bmod n+1)$.

For example, a delta-connected capacitor bank with admittance between phases $\mathbf{B}_m$, as

$$\mathbf{B}_m = b_m \cdot \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}, \qquad \mathbf{N}_h = \begin{bmatrix} a, b, c \end{bmatrix}. \tag{102}$$

A capacitor bank injects current

$$\mathbf{I}_m = j\,\mathbf{B}_m\,\mathbf{U}_i\,, \tag{103}$$

which is accounted for in the bus KCL (69) exactly as for a shunt (which would have $\mathbf{Y}_h = \mathbf{0} + j\mathbf{B}_m$).

3.7 Generators

Generators are modeled as a distinct entity from loads to account for power injections which are variable. Therefore, if a (real-world) generator has a fixed active and reactive power injection, the appropriate model within this specification is a load (with the active and reactive power values the negative value of the generated powers).

The current $\mathbf{I}_g$ represents the current sent *to* the bus i (i.e., in the opposite sense to a load). Generators observe conservation of current, i.e.

$$\mathbf{1}^{\mathrm{T}}\mathbf{I}_g = 0. \tag{104}$$

As for loads, set points are defined based on the phase-to-neutral voltages,

$$\mathbf{S}_g^{\perp} = \mathbf{P}_g^{\perp} + j\mathbf{Q}_g^{\perp} = \mathbf{U}_i^{\perp} \circ \mathbf{I}_g[\mathcal{P}]^* \tag{105}$$

Generators are dispatched between their lower $\mathbf{P}_g^{\min}$ and upper $\mathbf{P}_g^{\max}$ complex power bounds,

$$\mathbf{P}_g^{\min} \leq \mathbf{P}_g^{\perp} \leq \mathbf{P}_g^{\max}, \tag{106}$$

and between their lower $\mathbf{Q}_g^{\min}$ and upper $\mathbf{Q}_g^{\max}$ reactive power bounds

$$\mathbf{Q}_g^{\min} \leq \mathbf{Q}_g^{\perp} \leq \mathbf{Q}_g^{\max}. \tag{107}$$

Those dispatch values may also be constrained by current limits,

$$\mathbf{I}_g \circ (\mathbf{I}_g)^* \leq \mathbf{I}_g^{\max} \circ \mathbf{I}_g^{\max}. \tag{108}$$

Alternatively we can fix the generator set point,

$$\mathbf{S}_g^{\mathrm{ref},\perp} = \mathbf{S}_g^{\perp}, \tag{109}$$

which effectively becomes the same as a negative load.

We model linear costs for power generation by phase, in \$/kWh,

$$\mathbf{C}_g = \begin{bmatrix} C_{g,a} & C_{g,b} & C_{g,c} \end{bmatrix}^{\mathrm{T}}.$$

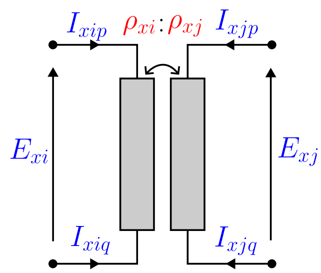

Figure 4: Idealized single-phase transformer circuit

3.8 Transformers

The mathematical model of transformers focuses on modeling of transformer turns ratios and connection of windings (through idealized transformer models), and series impedance modeling of copper losses and flux leakage. This model version does not support shunt impedance (typically used to models magnetizing reactance and eddy current losses) but will be considered in future versions.

The mathematical model for each transformer configuration (single-phase, three-phase delta-wye, and center-tap) vary qualitatively, meaning that within the data model they are each treated as different objects. Mathematically, these models can be considered as the connection of idealized pairs of transformer windings, Fig. 4, which are then combined into different configurations with series impedance.

The model of an idealized transformer is based on shared flux linkage between the pair of windings, conservation of complex power, and KCL. For winding EMF for the from, to E_{xi}, E_{xj} and number of turns ρ_{xi}, ρ_{xj} (both respectively), and winding currents as shown on Fig. 4, these constraints are, respectively,

$$\frac{E_{xi}}{\rho_{xi}} = \frac{E_{xj}}{\rho_{xj}}\,, \tag{110}$$

$$E_{xi}{I_{xip}}^* + E_{xj}{I_{xjp}}^* = 0\,, \tag{111}$$

$$I_{xip} + I_{xiq} = 0\,, \qquad I_{xjp} + I_{xjq} = 0\,. \tag{112}$$

It is most common for transformer nominal power to be defined, rather than current limits. The corresponding constraint on power transfer through the transformer is

$$|E_{xi}{I_{xip}}^*| \leq \frac{S_x^{\max}}{n_x}\,, \tag{113}$$

where $n_x = 1$ for a single-phase transformer, $n_x = 3$ for a three-phase transformer, whilst for a Center-Tap transformer $n_x = 1$ for the 'from' winding and $n_x = 2$ for the pair of 'to' windings (Figure 6).

Assuming non-zero values for all transformer variables, we can substitute (110) into (111) to yield

$$\rho_{xi} I_{xip} + \rho_{xj} I_{xjp} = 0, \tag{114}$$

The standard parameterization of a distribution transformer model is often provided in terms of the nominal voltage of its terminals, rather than the number of turns of each winding. (For a

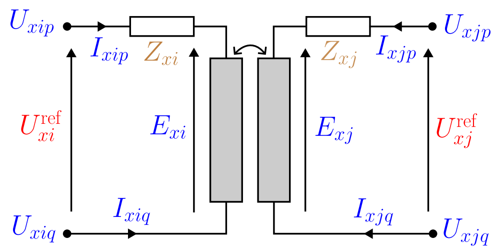

Figure 5: Single-phase transformer equivalent circuit

single-phase transformer such as this, this is unambiguous, as the nominal voltage is across the windings, U_{xi}^{ref}, U_{xj}^{ref}, but is more complex for multi-terminal transformers.) Assuming the ratio of nominal voltages U_{xi}^{ref}, U_{xj}^{ref} is in proportion to the turns ratio ρ_{xi}, ρ_{xj}, then (110), (111) can therefore equivalently be represented

$$\frac{E_{xi}}{U_{xi}^{\mathrm{ref}}} = \frac{E_{xj}}{U_{xj}^{\mathrm{ref}}}\,, \qquad U_{xi}^{\mathrm{ref}} I_{xip} + U_{xj}^{\mathrm{ref}} I_{xjp} = 0\,. \tag{115}$$

In distribution network applications, the side of the transformer with a higher nominal voltage rating is typically referred to as the Primary side, and the side of the transformer with a lower nominal voltage rating is referred to as the Secondary side.

3.8.1 Single-Phase Transformer

A circuit model for the single-phase transformer model is given in Fig. 5. The mathematical model for a single-phase transformer is found through the combinations of an idealized single-phase transformer (Section 3.8)

$$\frac{E_{xi}}{U_{xi}^{\mathrm{ref}}} = \frac{E_{xj}}{U_{xj}^{\mathrm{ref}}}\,, \tag{116}$$

$$U_{xi}^{\mathrm{ref}} I_{xip} + U_{xj}^{\mathrm{ref}} I_{xjp} = 0\,, \tag{117}$$

$$I_{xip} + I_{xiq} = 0\,, \tag{118}$$

$$I_{xjp} + I_{xjq} = 0\,, \tag{119}$$

and Ohm's law, as

$$E_{xi} = (U_{xip} - Z_{xi}^{\mathrm{s}} I_{xip}) - U_{xiq}\,, \tag{120}$$

$$E_{xj} = (U_{xjp} - Z_{xj}^{\mathrm{s}} I_{xjp}) - U_{xjq}\,. \tag{121}$$

3.8.2 Center-Tap Transformer

The circuit model of a center-tap transformer is shown in Fig. 6. As compared to a single-phase transformer, the 'centre tap' results in five terminals, two on the Primary (From) bus and three on the Secondary (To) bus.

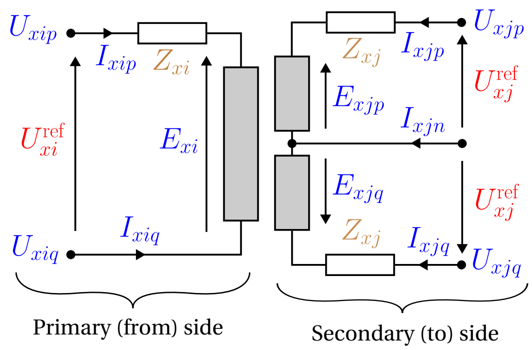

Figure 6: Center tap transformer equivalent circuit

The physical basis for the center-tap transformer is the same as the single-phase transformer. The impedance and nominal voltage rating parameters for each winding on the Secondary (To) side of the transformer are the same. Also note that the sign convention for the voltage across the upper and lower halves of the winding on the Secondary (To) side are in an opposite sense, with the central node connected to a (typically grounded) neutral wire.

First, considering KCL, the currents on each side of the transformer are

$$I_{xip} + I_{xiq} = 0\,, \qquad I_{xjp} + I_{xjn} + I_{xjq} = 0\,. \tag{122}$$

Ohm's law determines the voltages across the windings,

$$E_{xi} = (U_{xip} - Z^{\text{s}}_{xi} I_{xip}) - U_{xiq} \tag{123}$$
$$E_{xip} = (U_{xjp} - Z^{\text{s}}_{xj} I_{xjp}) - U_{xjn} \tag{124}$$
$$E_{xiq} = (U_{xjq} - Z^{\text{s}}_{xj} I_{xjq}) - U_{xjn}\,. \tag{125}$$

The idealized transformer component equations for flux linkage and complex power balance are

$$\frac{E_{xi}}{U^{\text{ref}}_{xi}} = \frac{E_{xip}}{U^{\text{ref}}_{xj}} = \frac{E_{xiq}}{U^{\text{ref}}_{xj}}\,, \tag{126}$$
$$U^{\text{ref}}_{xi} I_{xip} + U^{\text{ref}}_{xj} I_{xjp} + U^{\text{ref}}_{xj} I_{xjq} = 0\,. \tag{127}$$

3.8.3 Three-phase Wye-Delta and Delta-Wye Transformers

Figure 7 shows the connection of a Wye-Delta transformer. The mathematical model combines three single-phase transformers, with winding 1 electromagnetically linked to winding 4; winding 2 linked to winding 5; and winding 3 linked to winding 6.

KCL at each side of the transformer is

$$I_{xi,a} + I_{xi,b} + I_{xi,c} + I_{xi,n} = 0\,, \qquad I_{xj,a} + I_{xj,b} + I_{xj,c} = 0\,. \tag{128}$$

with the delta-side windings currents related to the terminal currents as

$$I_{xj,a} = I_{x,4} - I_{x,6}\,, \qquad I_{xj,b} = I_{x,5} - I_{x,4}\,, \qquad I_{xj,c} = I_{x,6} - I_{x,5}\,, \tag{129}$$

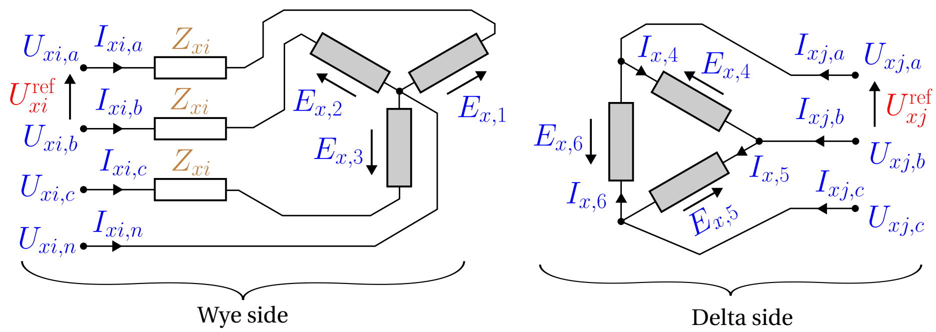

Figure 7: Three-phase wye-delta (or delta-wye) transformer equivalent circuit and variables. The reference voltages are the same for each side of the transformer (defining the turns ratio) for all pairs of phases $\{ab, bc, ca\}$. A Wye-Delta transformer has the Wye on the 'From' bus, whilst a Delta-Wye has the Delta on the 'From' bus.

noting that the constraints (128) implies that one of the three constraints of (129) is redundant (all three constraints are shown to illustrate symmetry). Ohm's law determines the voltages across the wye windings,

$$E_{x,1} = (U_{xi,a} - Z^{\mathrm{s}}_{xi} I_{xi,a}) - U_{xi,n}\,, \tag{130a}$$
$$E_{x,2} = (U_{xi,b} - Z^{\mathrm{s}}_{xi} I_{xi,b}) - U_{xi,n}\,, \tag{130b}$$
$$E_{x,3} = (U_{xi,c} - Z^{\mathrm{s}}_{xi} I_{xi,c}) - U_{xi,n}\,, \tag{130c}$$

and for the delta windings,

$$E_{x,4} = U_{xj,a} - U_{xj,b}\,, \qquad E_{x,5} = U_{xj,b} - U_{xj,c}\,, \qquad E_{x,6} = U_{xj,c} - U_{xj,a}\,. \tag{131}$$

The idealized transformer component equations for flux linkage are

$$\frac{\sqrt{3}E_{x,1}}{U^{\mathrm{ref}}_{xi}} = \frac{E_{x,4}}{U^{\mathrm{ref}}_{xj}}\,, \qquad \frac{\sqrt{3}E_{x,2}}{U^{\mathrm{ref}}_{xi}} = \frac{E_{x,5}}{U^{\mathrm{ref}}_{xj}}\,, \qquad \frac{\sqrt{3}E_{x,3}}{U^{\mathrm{ref}}_{xi}} = \frac{E_{x,6}}{U^{\mathrm{ref}}_{xj}}\,, \tag{132}$$

and, for complex power balance,

$$\frac{U^{\mathrm{ref}}_{xi}}{\sqrt{3}} I_{x,1} + U^{\mathrm{ref}}_{xj} I_{x,4} = 0\,, \qquad \frac{U^{\mathrm{ref}}_{xi}}{\sqrt{3}} I_{x,2} + U^{\mathrm{ref}}_{xj} I_{x,5} = 0\,, \qquad \frac{U^{\mathrm{ref}}_{xi}}{\sqrt{3}} I_{x,3} + U^{\mathrm{ref}}_{xj} I_{x,6} = 0\,, \tag{133}$$

noting that factor of $\sqrt{3}$ is because the reference voltage given on the Wye side of the transformer is phase-to-phase, whilst the winding's reference voltage is phase-to-neutral.

3.9 Voltage Sources

Voltage sources are used to represent the connection of a distribution system to the upstream (sub)transmission network or other sources of power, i.e., representing an 'external grid' providing power to the system. They typically also provide a voltage reference angle. The voltage source acts

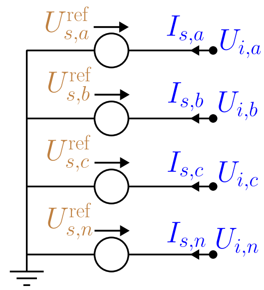

Figure 8: A four-terminal voltage source model, showing auxiliary currents $\mathbf{I}_s$ for modified nodal analysis.

to inject a voltage with respect to ground. Voltage sources are often combined with impedances to form a Thévenin equivalent.

A voltage source connected to bus i injects a voltage from the common ground to its other three terminals, such that

$$\mathbf{U}_i = \begin{bmatrix} U_{i,a} \\ U_{i,b} \\ U_{i,c} \\ U_{i,n} \end{bmatrix} = \begin{bmatrix} U_{s,a}^{\text{ref}} \\ U_{s,b}^{\text{ref}} \\ U_{s,c}^{\text{ref}} \\ U_{s,n}^{\text{ref}} \end{bmatrix} = \mathbf{U}_s^{\text{ref}}. \tag{134}$$

3.9.1 Incorporating Voltage Sources into an Optimization Problem

One can incorporate the voltage source information into the mathematical formulation in one of two ways (other approaches likely exist as well). In the first approach, we replace the nodal voltage variables in the nodal balance equations with the explicit values of the sources. In this approach, we do not include the nodal balance equations for the node at which the voltage source is connected (otherwise, the nodal balance set will not be full rank).

We follow the latter approach based on *modified nodal analysis* as the recommended option, as it is more generic and ideal for including models for independent voltage sources, switches and fuses (which can be modeled as voltage sources with zero-valued reference setpoint) without loss of generality. In this approach, we introduce additional equations to assign the voltage-source values, replacing (134) with

$$\mathbf{U}_i = \mathbf{U}_i^{\text{ref}}, \qquad \mathbf{I}_s = \begin{bmatrix} I_{s,a} \\ I_{s,b} \\ I_{s,c} \\ I_{s,n} \end{bmatrix}. \tag{135}$$

The additional equations introduces the auxiliary variable $\mathbf{I}_s$ (i.e., the current through the voltage source), as shown in Figure 8, and KCL for the currents drawn by the voltage source.

We next update the nodal balance equations at the voltage-source connections to include the auxiliary variable (i.e., the current through the voltage source).

$$\forall i: \underbrace{\sum_{lij \in \mathcal{T}^L} \mathbf{I}_{lij}}_{\text{lines}} + \underbrace{\sum_{xij \in \mathcal{T}^X} \mathbf{I}_{xij}}_{\text{transformers}} + \underbrace{\sum_{wij \in \mathcal{T}^W} \mathbf{I}_{wij}}_{\text{switches}} + \underbrace{\sum_{di \in \mathcal{C}^D} \mathbf{I}_d}_{\text{loads}} - \underbrace{\sum_{gi \in \mathcal{C}^G} \mathbf{I}_g}_{\text{generators}} + \underbrace{\sum_{hi \in \mathcal{C}^H} \mathbf{I}_h}_{\text{shunts}} + \underbrace{\sum_{si \in \mathcal{C}^S} \mathbf{I}_s}_{\text{voltage sources}} + \underbrace{\sum_{mi \in \mathcal{C}^K} \mathbf{I}_m}_{\text{capacitors}} = 0\,. \tag{136}$$

3.10 Switch

We use a switch object to represent sections with very low impedance relative to the rest of the network, e.g. switches, breakers, busbars or short lines. We utilize a lossless mathematical model for the switch.

The mathematical model for the switch is based on if the switch is open or closed. For a closed switch between bus i and j, the voltage at each bus must take the same value, and the current through each terminal of the switch is identical,

$$\mathbf{I}_{wij} + \mathbf{I}_{wji} = 0\,, \qquad \mathbf{U}_i = \mathbf{U}_j\,. \tag{137}$$

For an open switch, the current through the switch is zero, and the voltages are not linked through the switch,

$$\mathbf{I}_{wij} = \mathbf{I}_{wji} = 0\,, \quad \mathbf{U}_i, \mathbf{U}_j\,. \tag{138}$$

Current limits for a switch can be introduced as

$$\mathbf{I}_{wij} \circ (\mathbf{I}_{wij})^* \leq \mathbf{I}_w^{\max} \circ \mathbf{I}_w^{\max}\,. \tag{139}$$

3.11 Objective

We define an objective, using linear costs for power generation by phase,

$$\mathbf{C}_g = \begin{bmatrix} C_{g,a} & C_{g,b} & C_{g,c} \end{bmatrix}^{\mathrm{T}}.$$

Note that the costs are generally the same by phase, but we allow for these values to be different. The expression for the objective now becomes,

$$\min \sum_g (\mathbf{C}_g)^{\mathrm{T}} \Re(\mathbf{S}_g^{\perp}). \tag{140}$$

Note that we absolutely do *not* mean to imply that (linear) generation cost minimization is typically a realistic problem statement in distribution networks. However, it is well-posed optimization objective for these first efforts for benchmarking, as it can typically be expected to have a unique solution, as it is an equivalent problem to loss minimization. Future efforts can include alternative objective functions.

3.12 Summary of feasible set

Table 7 summarizes the feasible set of the optimization model. The variables listed enable voltages and currents within the network to be recovered for each of the individual elements. It is expected the full implementation of an optimization will include many of the variables described within the mathematical model, however, these will not necessarily be required to solve BMOPF problem of Table 7.

Objective	Generation cost		(140)
Bounds	Voltage magnitude	Phase-to-ground	(54) $\forall i \in \mathcal{I} \setminus \mathcal{I}^{\text{source}}$
		Phase-to-neutral	(57) $\forall i \in \mathcal{I} \setminus \mathcal{I}^{\text{source}}$
		Phase-to-phase	(60) $\forall i \in \mathcal{I} \setminus \mathcal{I}^{\text{source}}$
		Sequence	(65) $\forall i \in \mathcal{I} \setminus \mathcal{I}^{\text{source}}$
	Voltage angle difference	Bus pair	(67) $\forall ij \in \mathcal{B}$
		Phase-to-phase	(68) $\forall i \in \mathcal{I}$
	Current magnitude	Line	(74) $\forall lij \in \mathcal{T}^{L}$
		Switch	(139) $\forall wij \in \mathcal{T}^{W}$
		Generator	(108) $\forall g \in \mathcal{G}$
	Complex Power	Generator	(106), (107) $\forall g \in \mathcal{G}$
		Transformer	(113) $\forall xij \in \mathcal{T}^{X}$
Constraints	Lines	Ohm's law	(76), (77), (82) $\forall lij \in \mathcal{T}^{L\rightarrow}$
	Bus	KCL	(136) $\forall i \in \mathcal{I}$
	Generator, wye	power	(105) $\forall gi \in \mathcal{C}^{G}$
		power setpoint	(109) $\forall g \in \mathcal{G}$
		KCL	(104) $\forall g \in \mathcal{G}$
	Load, single-phase	power	(87) $\forall di \in (\mathcal{C}^{D})^{\Phi}$
		power setpoint	(87) $\forall d \in \mathcal{D}^{\Phi}$
		KCL	(86) $\forall d \in \mathcal{D}^{\Phi}$
	Load, wye	power	(89) $\forall di \in (\mathcal{C}^{D})^{⅄}$
		power setpoint	(90) $\forall d \in \mathcal{D}^{⅄}$
		KCL	(88) $\forall d \in \mathcal{D}^{⅄}$
	Load, delta	power	(93) $\forall di \in (\mathcal{C}^{D})^{\Delta}$
		power setpoint	(94) $\forall d \in \mathcal{D}^{\Delta}$
		KCL	(92) $\forall d \in \mathcal{D}^{\Delta}$
	Switch	closed	(137) $\forall wij \in \mathcal{T}^{W\rightarrow}$
		open	(138) $\forall wij \in \mathcal{T}^{W\rightarrow}$
	Voltage source	set point	(135) $\forall i \in \mathcal{I}^{\text{source}}$
	Transformer, ⅄Δ or Δ⅄	KCL	(128) $\forall x \in \mathcal{X}^{(⅄\Delta)} \cup \mathcal{X}^{(\Delta⅄)}$
		power balance	(133) $\forall x \in \mathcal{X}^{(⅄\Delta)} \cup \mathcal{X}^{(\Delta⅄)}$
		flux linkage	(132) $\forall x \in \mathcal{X}^{(⅄\Delta)} \cup \mathcal{X}^{(\Delta⅄)}$
		Ohm's law	(130) $\forall x \in \mathcal{X}^{(⅄\Delta)} \cup \mathcal{X}^{(\Delta⅄)}$
		Delta internal variables	(129), (131) $\forall x \in \mathcal{X}^{(⅄\Delta)} \cup \mathcal{X}^{(\Delta⅄)}$
	Transformer, single-phase	KCL	(118), (119) $\forall x \in \mathcal{X}^{\Phi}$
		power balance	(117) $\forall x \in \mathcal{X}^{\Phi}$
		flux linkage	(116) $\forall x \in \mathcal{X}^{\Phi}$
		Ohm's law	(120), (121) $\forall x \in \mathcal{X}^{\Phi}$
	Transformer, center-tap	KCL	(122) $\forall x \in \mathcal{X}^{\Upsilon}$
		power balance	(127) $\forall x \in \mathcal{X}^{\Upsilon}$
		flux linkage	(126) $\forall x \in \mathcal{X}^{\Upsilon}$
		Ohm's law	(123), (124), (125) $\forall x \in \mathcal{X}^{\Upsilon}$

Table 7: Feasible set of current-voltage UBOPF

4 Data Model Specification

To enable convenient data interchange, we define a data model and provide a schema [Work-in-progress]. The schema will provide basic checks that the data types for a model are correct. However, the schema only checks basic adherence to the data format. This document provides the most complete definition of the data model and admissible values.

4.1 Data Input Formatting

4.1.1 Matrices and Complex Numbers

The mathematical model (Section 3) defines a vectors, matrices in both complex and real numbers. JSON supports ordered lists of real numbers. To represent complex numbers, vectors and matrices, the complex-valued object is represented as a pair of real-valued objects, either as rectangular or polar representations.

To represent matrices, we use row-first ordering. For example, the a real-valued matrix $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots \\ \vdots & \ddots & \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \tag{141}$$

would be represented in JSON format (1-indexed) with an underscore delimiter, as

- `A` $\rightarrow$
  - `A_1_1` $\rightarrow a_{11}$
  - `A_1_2` $\rightarrow a_{12}$
  - ...
  - `A_m_n` $\rightarrow a_{mn}$

4.1.2 Data Model Units

The data model does not include any entries to define units of entered values; instead, the units in the data model are shown in Table 8. For basic physical measured quantities, SI Units are used. The non-SI unit of energy, kWh is equivalent to the energy that would be used if a 1 kW load was used for an hour (i.e., 3.6×10^6 J). This is used as a part of the cost rate quantity, \$/kWh, due to its widespread use in industry and academia (noting there is no SI quantity of currency).

Three examples of the conversion from conventional power systems representation to the JSON format in SI Units are given in Table 9. To enable accurate comparison with tools with non-SI units (for example, converting between radians and degrees), it is preferred to give the value of constants to a minimum of 10 significant figures, and ideally to full floating point precision.

4.1.3 Grounding with the Data Model

Within both the data and mathematical models, ground is treated as a 'copper plate', 0 V reference across the entire network. As shown in Table 10, the data model assumes that most elements are not connected to ground. Instead, elements are connected to a terminal at a bus which is then connected to ground. This means that KCL can be applied at the terminal of most elements.

Quantity	Unit	Symbol
Voltage	Volt	(V)
Current	Ampere	(A)
Length	Meter	(m)
(Active) Power	Watt	(W)
Reactive power	Volt-ampere reactive	(var)
Apparent power	Volt-ampere	(VA)
Conductance	Siemens	(S)
Susceptance	Siemens	(S)
Resistance	Ohm	(Ω)
Reactance	Ohm	(Ω)
Angle	Radians	(rad)
Cost rate	USD per kW-hour	($/kWh)

Table 8: Units used in the Data Model

Quantity	Conventional power systems unit	SI Unit	Example	
			Conventional	JSON format
Active Power	Kilowatts (kW)	Watts (W)	3 kW	`3e3` (or `3000`, or `3.0e3`)
Angle	Degrees	Radians	120°	`1.0471975511965976` (or `1.0471975511965976e0`)

Table 9: Converting from conventional power systems units to SI units and their representation within a JSON file.

Type	Has connection to ground			Comment
	Never	Optional	Always	
Load	✓			Grounding cast as bus property or shunt.
Generator	✓			Grounding cast as bus property or shunt.
Transformers	✓			Grounding cast as bus property or shunt.
Capacitor	✓			Grounding cast as bus property or shunt.
Switch	✓			Switches are not usually grounded.
Bus terminals		✓		A flag allows for perfect grounding of a bus's terminal(s). NB: grounding through an impedance is via a Shunt.
Line			✓	If a non-zero branch shunt admittance is defined, current will flow in the ground
Shunt			✓	Grounding a conductor through an impedance is via a shunt (perfect grounding is a bus terminal's property).
Voltage source			✓	Voltage sources are defined voltage-to-ground.

Table 10: Applicability of element grounding within the data model

Example naming convention	JSON bus terminal names $\mathbf{N}_i$
Bus A in Figure 1	`["a","b","c","n"]`
Bus D in Figure 1	`["a","b","c"]`
Four-node bus in OpenDSS	`["1","2","3","4"]`
Triplex bus in Gridlab-D	`["1","N","2"]`
ABC or ABCD bus in Gridlab-D	`["A","B","C","N"]`
IEC 60445 conductor alphanumeric notation	`["L1","L2","L3","N"]`

Table 11: Example conversion from different tools JSON bus terminal names $\mathbf{N}_i$.

There are two exceptions to this rule. The first are Lines, for which the line shunt impedance is defined with respect to ground, so lines will therefore always have an implicit connection to ground. A Shunt will also always have an implicit connection to ground. (These implicit connections are illustrated in Figure 1.)

4.1.4 Permissible Strings and String Arrays

To enable various conventions for conductors names at a bus to be captured, terminals have string type. Table 11 provides some examples of permissible terminal names seen across tools and in standards. The list of restrictions for other string-valued parameters used in the data model are given in Table 12.

Symbol	Represents	Value restrictions within JSON format
$\mathbf{N}_i$	Name given to each terminal of a bus	None
T_d	Load configuration	Configuration (Table 4)
T_g	Generator configuration	"
T_m	Capacitor configuration	"
$\mathbf{G}_i$	Array of perfectly grounded terminals at a bus	Array entries must be within corresponding $\mathbf{N}_i$
$\mathbf{N}_s$	Voltage source to bus terminal mapping	"
$\mathbf{N}_h$	Shunt to bus terminal mapping	"
$\mathbf{N}_d$	Load to bus terminal mapping	"
$\mathbf{N}_g$	Generator to bus terminal mapping	"
$\mathbf{N}_m$	Capacitor to bus terminal mapping	"
$\mathbf{N}_{li}$, $\mathbf{N}_{lj}$	Line to bus terminal mappings	"
$\mathbf{N}_{wi}$, $\mathbf{N}_{wj}$	Switch to bus terminal mappings	"
$\mathbf{N}_{xi}$, $\mathbf{N}_{xj}$	Transformer to bus terminal mapping	"
m_d	Load model types	`CONSTANT_POWER`, `CONSTANT_CURRENT`, `CONSTANT_IMPEDANCE`, `ZIP`, `EXPONENTIAL`

Table 12: String-valued parameters

4.1.5 Required and Optional Fields within the Data Model

Each of the element types of the data model has some required fields. However, for some of the element types, some fields are optional. The interpretation of these optional fields varies.

- If an optional field for a constraint is not filled within a model, it implies that the corresponding constraint is unbounded. For example, if a voltage magnitude upper bound is not provided, it implies that there is no upper bound on the voltage at that bus.
- If an optional field for a parameter is not passed, it implies a null value should be considered for that parameter. For example, if a value for the series resistance of the 'from' side of a center-tap transformer is not given in a given network model, it implies the resistance between the 'from' winding and the transformer's 'from' terminals has value 0 Ω.

4.2 Document Metadata (`meta`)

Every BMOPF document *should* carry a top-level `meta` object providing provenance, licensing, and versioning information. Conformant writers *must* populate `meta` when generating a document; readers *must* tolerate its absence for compatibility with documents produced before this field was introduced. The object is flat except for three structured sub-objects: `authors`, `sources`, and `generator`.

4.2.1 Top-level `meta` fields

Table 13 defines the common fields that describe the document as a whole, including its schema, dataset version, provenance, and licensing information. The structured `authors`, `sources`, and `generator` fields are summarized here and specified individually in the following subsections.

Table 13: Fields of the `meta` object. All fields are optional; conformant writers should populate at minimum `$schema`, `created`, and `license`.

Field	Type	Description
`$schema`	string (URI)	URI of the BMOPF JSON Schema document against which this file was validated. Enables schema-aware tools to locate and apply the correct validator without out-of-band configuration.
`version`	string	Version of this *dataset* (not the schema). Allows publishers to distinguish successive releases of the same network (e.g. corrected impedances, extended time series).
`title`	string	Human-readable name for the dataset. Distinct from the machine-readable root-level `name` field, which is used as a stable identifier by analysis tools.
`description`	string	Free-text description, modelling notes, or caveats.
`created`	string (ISO 8601)	Date and time at which this document was generated, in UTC, e.g. `2026-06-15T10:00:00Z`.
`modified`	string (ISO 8601)	Date and time of the most recent revision, in UTC.
`license`	string	Licence under which the dataset may be used. Either an SPDX licence identifier (e.g. `CC-BY-4.0`) or a `https://` URI pointing to the full licence text.
`authors`	array of objects	Ordered list of persons or organisations responsible for the dataset. See Table 14.
`sources`	array of objects	References to the original data from which this document was derived. See Table 15.
`generator`	object	Software tool that produced this document. See Table 16.

4.2.2 Author object (`meta.authors[]`)

Each entry in `meta.authors` identifies a person or organisation responsible for preparing or publishing the dataset. Table 14 lists the available attribution and contact fields; the array order should preserve the publisher's intended author ordering.

Table 14: Fields of each entry in `meta.authors`.

Field	Type	Description
`name`	string	Full name of the author or organisation.
`email`	string	Contact e-mail address.
`orcid`	string	ORCID persistent identifier in the canonical hyphenated form `XXXX-XXXX-XXXX-XXXX`, where the last character may be a digit or the letter `X`.

4.2.3 Source object (`meta.sources[]`)

The `sources` array records the lineage of the engineering data rather than authorship of the BMOPF document itself. Table 15 provides complementary identifiers for each source so that users can locate the original data and distinguish independently versioned inputs.

Table 15: Fields of each entry in `meta.sources`.

Field	Type	Description
`name`	string	Human-readable name of the source dataset or file.
`url`	string (URI)	Dereferenceable `https://` URI where the source data can be retrieved.
`format`	string	Format of the source data, e.g. `OpenDSS`, `PMD-JSON`, `MATPOWER`.
`doi`	string	Digital Object Identifier of an associated publication or data repository entry, e.g. `10.1145/3538637.3538844`.
`version`	string	Version string of the source dataset, if versioned independently.

4.2.4 Generator object (`meta.generator`)

The `generator` object identifies the software responsible for serialising the current BMOPF document. Its fields, listed in Table 16, distinguish the producing tool's version from both the dataset version and the BMOPF schema version.

4.2.5 Example

The following example shows how the top-level metadata fields and their three structured sub-objects are assembled in a complete BMOPF document. The values are illustrative, but the example also demonstrates the intended separation between the root-level machine identifier and human-readable metadata.

Table 16: Fields of `meta.generator`.

Field	Type	Description
`tool`	string	Name of the software tool or library that wrote this file.
`version`	string	Version of that tool, supporting reproducibility audits.


```
{
  "name": "enwl_network1_feeder1",
  "meta": {
    "$schema": "https://example.org/bmopf/schema/v1/bmopf.json",
    "version": "1.0",
    "title": "ENWL LV Network 1, Feeder 1",
    "description": "Four-wire 230/400 V LV feeder derived from the ENWL
                    Low Voltage Network Solutions dataset.",
    "created": "2026-06-15T10:00:00Z",
    "license": "https://creativecommons.org/licenses/by/4.0/",
    "authors": [
      {
        "name": "Frederik Geth",
        "email": "frederik.geth@example.org",
        "orcid": "0000-0001-9534-2265"
      }
    ],
    "sources": [
      {
        "name": "ENWL Low Voltage Network Solutions dataset",
        "url":  "https://www.enwl.co.uk/lvns",
        "format": "OpenDSS",
        "doi":  "10.1145/3538637.3538844"
      }
    ],
    "generator": {
      "tool":    "BMOPFTools.jl",
      "version": "0.1.0"
    }
  },
  "bus": { ... },
  "line": { ... }
}
```

4.2.6 Conformance notes

The following notes refine the table definitions where interoperability depends on a particular lexical form or reader behaviour. They should be applied together with Tables 13–16 when validating or generating the `meta` object.

- The `$schema` field uses the JSON Schema `$schema` keyword convention [?]. Its value *should* be a versioned, dereferenceable URI; a floating reference to `main` is acceptable during the draft period.
- Dates *must* be UTC and *should* include a full time component (`THH:MM:SSZ`). Date-only values (`YYYY-MM-DD`) are permitted where the time of day is unknown.
- ORCID values *must* omit the `https://orcid.org/` prefix; the bare hyphenated identifier is the canonical form.
- The `license` field *should* be a URI when the licence requires specific attribution language not captured by an SPDX identifier alone.
- Additional fields not listed above *may* be present; conformant readers *must* ignore unknown fields.

4.3 Terminal conventions

Terminal identifiers are *strings*. Each bus lists its conductors in an ordered array `terminal_names`; component identifiers themselves carry no ordering semantics. The canonical written form is `"1", "2", "3", "n"`, but the following naming conventions are recognised on read:

Convention	Phases	Neutral
OpenDSS numeric	`"1" "2" "3"`	`"4"` (positional)
Canonical	`"1" "2" "3"`	`"n"`
Letter	`"a" "b" "c"`	`"n"/"N"`
IEC 60445	`"L1" "L2" "L3"`	`"N"`

Table 17: Recognised terminal-naming conventions (case-insensitive).

Terminal roles. Because the label alone does not say which conductor is a phase or a neutral, a case may classify roles *once* for the whole network in a top-level `terminal_conventions` block:

```
"terminal_conventions": { "phase": ["a","b","c"], "neutral": ["n"]}
```

When present the block is **authoritative** (labels matched exactly, including case) and is always emitted on export. Role lists must be disjoint; the ground *reference* stays implicit. When the block is absent, roles are *inferred* (a terminal `n/N` is the neutral; terminal `"4"` when the bus terminal set is exactly $\{1, 2, 3, 4\}$; everything else a phase).

4.4 Full Data Model

The nested data model is organized as follows, with all IDs indexing the sets as strings (e.g., index i for a `bus` could take value `"1"`, noting that these indices are *not* ordered). Optional fields are indicated by the text *Optional:* being prepended to the field.

- `name` $\rightarrow$ "<network_name>"
- `bus` $\rightarrow$
  - $i \in \mathcal{I} \rightarrow$
    - `terminal_names` $\rightarrow \mathbf{N}_i$
    - *Optional:* `perfectly_grounded_terminals` $\rightarrow \mathbf{G}_i$
    - *Optional:* `v_min` $\rightarrow \mathbf{U}_i^{\text{min}}$ (V)
    - *Optional:* `v_max` $\rightarrow \mathbf{U}_i^{\text{max}}$ (V)
    - *Optional:* `vpn_min` $\rightarrow \mathbf{U}_i^{\perp,\text{min}}$ (V)– can be used for single-phase ($\mathbf{U}_i^{\perp,\text{min}}$ is length-1) triplex ($\mathbf{U}_i^{\perp,\text{min}}$ is length-2) and four-wire ($\mathbf{U}_i^{\perp,\text{min}}$ is length-3).
    - *Optional:* `vpn_max` $\rightarrow \mathbf{U}_i^{\perp,\text{max}}$ (V)- can be used for single-phase ($\mathbf{U}_i^{\perp,\text{max}}$ is length-1) triplex ($\mathbf{U}_i^{\perp,\text{max}}$ is length-2) and four-wire ($\mathbf{U}_i^{\perp,\text{max}}$ is length-3)
    - *Optional:* `vpp_min` $\rightarrow \mathbf{U}_i^{\Delta,\text{min}}$ (V)–only meaningful if $|\mathbf{N}_i| \geq 3$
    - *Optional:* `vpp_max` $\rightarrow \mathbf{U}_i^{\Delta,\text{max}}$ (V)–only meaningful if $|\mathbf{N}_i| \geq 3$
    - *Optional:* `vpos_min` $\rightarrow U_{i,1}^{\text{max}}$ (V)–only meaningful if $|\mathbf{N}_i| \geq 3$
    - *Optional:* `vpos_max` $\rightarrow U_{i,1}^{\text{max}}$ (V)–only meaningful if $|\mathbf{N}_i| \geq 3$
    - *Optional:* `vneg_max` $\rightarrow U_{i,2}^{\text{max}}$ (V)–only meaningful if $|\mathbf{N}_i| \geq 3$
    - *Optional:* `vzero_max` $\rightarrow U_{i,0}^{\text{max}}$ (V)–only meaningful if $|\mathbf{N}_i| \geq 3$
  - $j \in \mathcal{I} \rightarrow$
    - ...
- `line` $\rightarrow$
  - $l \in \mathcal{L} \rightarrow$
    - *Optional (linecode definition)* `length` $\rightarrow \ell_l$ (m)
    - *Optional (linecode definition)* `linecode` $\rightarrow c$
    - `bus_from` $\rightarrow i$
    - `bus_to` $\rightarrow j$
    - `terminal_map_from` $\rightarrow \mathbf{N}_{li}$
    - `terminal_map_to` $\rightarrow \mathbf{N}_{lj}$
    - *Optional (impedance definition):* `R_series` $\rightarrow \Re(\mathbf{Z}_l^{\text{s}})$ (Ω)–matrix entry
    - *Optional (impedance definition):* `X_series` $\rightarrow \Im(\mathbf{Z}_l^{\text{s}})$ (Ω)–matrix entry
    - *Optional (impedance definition):* `G_from` $\rightarrow \Re(\mathbf{Y}_{lij}^{\text{sh}})$ (S)–matrix entry
    - *Optional (impedance definition):* `G_to` $\rightarrow \Re(\mathbf{Y}_{lji}^{\text{sh}})$ (S)–matrix entry
    - *Optional (impedance definition):* `B_from` $\rightarrow \Im(\mathbf{Y}_{lij}^{\text{sh}})$ (S)–matrix entry
    - *Optional (impedance definition):* `B_to` $\rightarrow \Im(\mathbf{Y}_{lji}^{\text{sh}})$ (S)–matrix entry

    - *Optional (impedance definition):* `i_max` $\rightarrow \mathbf{I}_{lij}^{\max}$ (A)
    - *Optional (impedance definition):* `s_max` $\rightarrow \mathbf{S}_{lij}^{\max}$ (VA)

- `linecode` $\rightarrow$
  - $c \in \mathcal{C} \rightarrow$
    - `R_series` $\rightarrow \Re(\mathbf{Z}_c^{\mathrm{s}})$ (Ω/m)–matrix entry
    - `X_series` $\rightarrow \Im(\mathbf{Z}_c^{\mathrm{s}})$ (Ω/m)–matrix entry
    - *Optional:* `G_from` $\rightarrow \Re(\mathbf{Y}_c^{\mathrm{sh}})$ (S/m)–matrix entry
    - *Optional:* `G_to` $\rightarrow \Re(\mathbf{Y}_c^{\mathrm{sh}})$ (S/m)–matrix entry
    - *Optional:* `B_from` $\rightarrow \Im(\mathbf{Y}_c^{\mathrm{sh}})$ (S/m)–matrix entry
    - *Optional:* `B_to` $\rightarrow \Im(\mathbf{Y}_c^{\mathrm{sh}})$ (S/m)–matrix entry
    - *Optional:* `i_max` $\rightarrow \mathbf{I}_{lij}^{\max}$ (A)
    - *Optional:* `s_max` $\rightarrow \mathbf{S}_{lij}^{\max}$ (VA)
    - *Optional:* `source` $\rightarrow$ string indicating data source

- `voltage_source` $\rightarrow$
  - $s \in \mathcal{S} \rightarrow$
    - `v_magnitude` $\rightarrow |\mathbf{U}_s^{\mathrm{ref}}|$ (V)
    - `v_angle` $\rightarrow \angle\mathbf{U}_s^{\mathrm{ref}}$ (radians)
    - `bus` $\rightarrow i$
    - `terminal_map` $\rightarrow \mathbf{N}_s$

- `generator` $\rightarrow$
  - $g \in \mathcal{G} \rightarrow$
    - `bus` $\rightarrow i$
    - `terminal_map` $\rightarrow \mathbf{N}_g$
    - `configuration` $\rightarrow T_d$
    - `cost` $\rightarrow \mathbf{C}_g$ (\$/kWh)
    - *Optional:* `p_min` $\rightarrow \mathbf{P}_g^{\min}$ (W)
    - *Optional:* `p_max` $\rightarrow \mathbf{P}_g^{\max}$ (W)
    - *Optional:* `q_min` $\rightarrow \mathbf{Q}_g^{\min}$ (var)
    - *Optional:* `q_max` $\rightarrow \mathbf{Q}_g^{\max}$ (var)

- `load` $\rightarrow$
  - $d \in \mathcal{D} \rightarrow$
    - `p_nom` $\rightarrow \mathbf{P}_d^{\mathrm{ref},\,(\curlywedge\,\mathrm{or}\,\Delta\,\mathrm{or}\,\Phi)}$ (W)
    - `q_nom` $\rightarrow \mathbf{Q}_d^{\mathrm{ref},\,(\curlywedge\,\mathrm{or}\,\Delta\,\mathrm{or}\,\Phi)}$ (var)
    - `bus` $\rightarrow j$
    - `terminal_map` $\rightarrow \mathbf{N}_d$
    - `configuration` $\rightarrow T_d$

        - *Optional:* `model` $\rightarrow m_d$
        - *Optional (ZIP model):* `v_nom` $\rightarrow U_d^{\text{nom}}$ (V)
        - *Optional (ZIP model):* `alpha_z` $\rightarrow \alpha_{d,k}^{Z}$
        - *Optional (ZIP model):*`alpha_i` $\rightarrow \alpha_{d,k}^{I}$,
        - *Optional (ZIP model):* `alpha_p` $\rightarrow \alpha_{d,k}^{P}$
        - *Optional (ZIP model):* `beta_z` $\rightarrow \beta_{d,k}^{Z}$
        - *Optional (ZIP model):* `beta_i` $\rightarrow \beta_{d,k}^{I}$
        - *Optional (ZIP model):* `beta_p` $\rightarrow \beta_{d,k}^{P}$
        - *Optional (exponential model):* `gamma_p` $\rightarrow \gamma_{d,k}^{P}$,
        - *Optional (exponential model):* `gamma_q` $\rightarrow \gamma_{d,k}^{Q}$
    - `shunt` $\rightarrow$
- $h \in \mathcal{H} \rightarrow$
    - `G` $\rightarrow \Re(\mathbf{Y}_h)$ (S)–matrix entry
    - `B` $\rightarrow \Im(\mathbf{Y}_h)$ (S)–matrix entry
    - `bus` $\rightarrow j$
    - `terminal_map` $\rightarrow \mathbf{N}_h$
- `capacitor` $\rightarrow$
    - $m \in \mathcal{K} \rightarrow$
        - `q_nom` $\rightarrow Q_m^{\text{nom}}$ (var)
        - `v_nom` $\rightarrow U_m^{\text{nom}}$ (V)
        - `bus` $\rightarrow j$
        - `terminal_map` $\rightarrow \mathbf{N}_m$
        - `configuration` $\rightarrow T_m$
- `switch` $\rightarrow$
    - $w \in \mathcal{W} \rightarrow$
        - `bus_from` $\rightarrow i$
        - `bus_to` $\rightarrow j$
        - `open_switch` $\rightarrow$ `true` or `false`
        - `terminal_map_from` $\rightarrow \mathbf{N}_{wi}$
        - `terminal_map_to` $\rightarrow \mathbf{N}_{wj}$
        - *Optional:* `i_max` $\rightarrow \mathbf{I}_w^{\text{max}}$ (A)
- `transformer` $\rightarrow$
    - `single_phase` $\rightarrow$
        - $x \in \mathcal{X}^{\Phi} \rightarrow$
            - `bus_from` $\rightarrow i$

- · `terminal_map_from` $\to \mathbf{N}_{xi}$ (length 2)
- · `v_ref_from` $\to U^{\text{ref}}_{xi}$ (V)
- · `bus_to` $\to j$
- · `terminal_map_to` $\to \mathbf{N}_{xj}$ (length 2)
- · `v_ref_to` $\to U^{\text{ref}}_{xj}$ (V)
- · `s_rating` $\to S^{\max}_{x}$ (VA)
- · *Optional:* `r_series_from` $\to \Re(Z^{\text{s}}_{xi})$ (Ω)
- · *Optional:* `x_series_from` $\to \Im(Z^{\text{s}}_{xi})$ (Ω)
- · *Optional:* `r_series_to` $\to \Re(Z^{\text{s}}_{xj})$ (Ω)
- · *Optional:* `x_series_to` $\to \Im(Z^{\text{s}}_{xj})$ (Ω)

– `wye_delta` $\to$
  * $x \in \mathcal{X}^{\curlywedge\Delta} \to$
    - · `bus_from` $\to i$
    - · `terminal_map_from` $\to \mathbf{N}_{xi}$ (length 4)
    - · `v_ref_from` $\to U^{\text{ref}}_{xi}$ (V)
    - · `bus_to` $\to j$
    - · `terminal_map_to` $\to \mathbf{N}_{xj}$ (length 3)
    - · `v_ref_to` $\to U^{\text{ref}}_{xj}$ (V)
    - · `s_rating` $\to S^{\max}_{x}$ (VA)
    - · *Optional:* `r_series` $\to \Re(Z^{\text{s}}_{x})$ (Ω)
    - · *Optional:* `x_series` $\to \Im(Z^{\text{s}}_{x})$ (Ω)

– `delta_wye` $\to$
  * $x \in \mathcal{X}^{\Delta\curlywedge} \to$
    - · `bus_from` $\to i$
    - · `terminal_map_from` $\to \mathbf{N}_{xi}$ (length 3)
    - · `v_ref_from` $\to U^{\text{ref}}_{xi}$ (V)
    - · `bus_to` $\to j$
    - · `terminal_map_to` $\to \mathbf{N}_{xj}$ (length 4)
    - · `v_ref_to` $\to U^{\text{ref}}_{xj}$ (V)
    - · `s_rating` $\to S^{\max}_{x}$ (VA)
    - · *Optional:* `r_series` $\to \Re(Z^{\text{s}}_{x})$ (Ω)
    - · *Optional:* `x_series` $\to \Im(Z^{\text{s}}_{x})$ (Ω)

– `center_tap` $\to$
  * $x \in \mathcal{X} \to$
    - · `bus_from` $\to i$
    - · `terminal_map_from` $\to \mathbf{N}_{xi}$ (length 2)
    - · `v_ref_from` $\to U^{\text{ref}}_{xi}$ (V)
    - · `bus_to` $\to j$
    - · `terminal_map_to` $\to \mathbf{N}_{xj}$ (length 3)

- `v_ref_to` → U^{ref}_{xj} (V)
- `s_rating` → $S^{\max}_{x}$ (VA)
- *Optional:* `r_series_from` → $\Re(Z^{\text{s}}_{xi})$ (Ω)
- *Optional:* `x_series_from` → $\Im(Z^{\text{s}}_{xi})$ (Ω)
- *Optional:* `r_series_to` → $\Re(Z^{\text{s}}_{xj})$ (Ω)
- *Optional:* `x_series_to` → $\Im(Z^{\text{s}}_{xj})$ (Ω)

Appendix A

A.1 Frequently Asked Questions

How can I define a triplex load (as defined in `Gridlab-D`)? A triplex load can be defined as one or more single phase loads. For example, to implement a 1 kW triplex load connected across phases '1' to '2', a 'Triplex' bus can be defined with terminal names `["1","n","2"]`, then a 1 kW single phase load defined with terminal mapping `["1","2"]`.

I have converted a 'Wye' load to a 'Delta' load using the well-known delta-wye transformation. Why do I get a different problem solution? The Wye load in power distribution engineering is a Wye load with neutral return conductor (four wires), as compared to a Wye load in other electrical engineering contexts or graph theory (which has three wires, without a return conductor). Therefore, the well-known delta-wye transformation is not applicable to Wye type loads, which have a midpoint return (Table 4).

If you wish to implement a three-wire Wye without return you can (i) add an additional 'floating' midpoint node terminal at the bus the Wye without midpoint return is implemented, (ii) add three single-phase loads that connect across the relevant phases to that midpoint node at that bus (or, a Wye with return with the neutral connected to that floating node). At present there are no plans to implement a specific load subclass for a Wye without midpoint return.

How can I calculate Center-Tap Transformer impedances values from `OpenDSS`-style impedance parameters? `OpenDSS` represents center-tap impedances through `Xhl`, `Xlt`, and `Xht` parameters, given in % per-unit values. These are linked to per-unit impedance values on the primary and each of the secondary windings `Xh`, `Xl`, `Xt` (respectively) through

$$\begin{bmatrix}\texttt{Xhl}\\ \texttt{Xlt}\\ \texttt{Xht}\end{bmatrix} = \begin{bmatrix}1 & 1 & 0\\ 0 & 1 & 1\\ 1 & 0 & 1\end{bmatrix}\begin{bmatrix}\texttt{Xh}\\ \texttt{Xl}\\ \texttt{Xt}\end{bmatrix} \iff \begin{bmatrix}\texttt{Xh}\\ \texttt{Xl}\\ \texttt{Xt}\end{bmatrix} = \frac{1}{2}\begin{bmatrix}1 & -1 & 1\\ 1 & 1 & -1\\ -1 & 1 & 1\end{bmatrix}\begin{bmatrix}\texttt{Xhl}\\ \texttt{Xlt}\\ \texttt{Xht}\end{bmatrix}. \tag{142}$$

The transformation from per-unit to Ω depends on the power base. The impedance base is implied by the `OpenDSS` parameters `kvs`, `kVAs`, the voltage and power rating of the windings of the transformer. The impedance values for the primary and secondary of the transformer (in Ω) are given in terms of these parameters as

$$Z^{\text{s}}_{xi} = \frac{\texttt{Xh}}{100} \times \frac{(1000 \times \texttt{kv[h]})^2}{(1000 \times \texttt{kva[h]})}, \qquad Z^{\text{s}}_{xj} = \frac{\texttt{Xl}}{100} \times \frac{(1000 \times \texttt{kv[l]})^2}{(1000 \times \texttt{kva[l]})}. \tag{143}$$

How can I calculate Center-Tap Transformer impedances values from `Gridlab-D` Center Tap transformer parameters? Basic parameters for assigning Gridlab-D Center-Tap transformer impedances are primary `impedance` (in (pu)), secondary `impedance1` (in (pu)), `primary_voltage` (in (V)), `secondary_voltage` (in (V), from phase-to-neutral), and `power_rating` (in (kVA), for the *total* transformer power). The Center-Tap transformer impedances in SI units can be calculated as

$$Z^{\mathrm{s}}_{xi} = \texttt{impedance} \times \frac{3}{1000} \times \frac{\texttt{primary\_voltage}^2}{\texttt{power\_rating}}\,, \tag{144}$$

$$Z^{\mathrm{s}}_{xj} = \times \texttt{impedance1} \times \frac{3}{1000} \times \frac{\texttt{secondary\_voltage}^2}{\texttt{power\_rating}}\,. \tag{145}$$

The factor of 1000 is required to converter the power rating from kVA to VA; the factor of 3 is because the total transformer `power_rating` is three times the power of the primary winding (in the absence of further per-winding parameter specification within the GRIDLAB-D model).

I have a distribution model defined line impedances represented as sequence components only. Can this be converted to wire coordinates? Sequence components can be converted to (possibly Kron-reduced) wire coordinates through the Fortescue transformation. The transformation required to convert from a positive and zero sequence impedance to full impedance matrix is detailed in Section A.2.

Why do we need another data model? There are a couple of gaps we are trying to fill with this initiative. We define 1) the semantics of our data model, 2) we design it in a way to enable better data quality checks, 3) we focus on the research applications related to optimization models. Down the line this work can lead to an extension of the CIM.

A.2 Calculating Line Impedance Matrices Given Symmetrical Components

By considering the sequence transformation matrix $\mathbf{F}$, sequence currents and voltages are linked to phase voltages and currents as

$$\mathbf{U}^{012}_{i} = \mathbf{F}\mathbf{U}_{i}[\mathcal{P}]\,, \tag{146}$$

$$\mathbf{I}^{012,\mathrm{s}}_{lij} = \mathbf{F}\mathbf{I}^{\mathrm{s}}_{lij}[\mathcal{P}]\,, \tag{147}$$

Ohm's law is written

$$\mathbf{U}_{j}[\mathcal{P}] = \mathbf{U}_{i}[\mathcal{P}] - \mathbf{Z}^{\mathrm{s}}_{l}\mathbf{I}^{\mathrm{s}}_{lij}[\mathcal{P}]\,, \tag{148}$$

where, for the purpose of considering sequence components, $\mathbf{Z}^{\mathrm{s}}_{l}$ is either true impedance of a three-wire system, or a Kron-reduced four wire system (either of which are a 3×3 matrix). Premultiplying by the matrix $\mathbf{F}$ and, considering the identity $I = \mathbf{F}^{-1}\mathbf{F}$ yields

$$\mathbf{F}\mathbf{U}_{j}[\mathcal{P}] = \mathbf{F}\mathbf{U}_{i}[\mathcal{P}] - \underbrace{\mathbf{F}\mathbf{Z}^{\mathrm{s}}_{l}\mathbf{F}^{-1}}_{\mathbf{Z}^{012,\mathrm{s}}_{l}}\mathbf{F}\mathbf{I}^{\mathrm{s}}_{lij}[\mathcal{P}]\,, \tag{149}$$

$$\mathbf{U}^{012}_{j} = \mathbf{U}^{012}_{i} - \mathbf{Z}^{012,\mathrm{s}}_{l}\mathbf{I}^{012,\mathrm{s}}_{lij}\,, \tag{150}$$

so,

$$\mathbf{Z}_l^{012,\mathrm{s}} = \mathbf{F}\mathbf{Z}_l^{\mathrm{s}}\mathbf{F}^{-1} = \begin{bmatrix} Z_{l,00}^{\mathrm{s}} & Z_{l,01}^{\mathrm{s}} & Z_{l,02}^{\mathrm{s}} \\ Z_{l,10}^{\mathrm{s}} & Z_{l,11}^{\mathrm{s}} & Z_{l,12}^{\mathrm{s}} \\ Z_{l,20}^{\mathrm{s}} & Z_{l,21}^{\mathrm{s}} & Z_{l,22}^{\mathrm{s}} \end{bmatrix}. \tag{151}$$

It is common to assume transposition of lines, so that off-diagonal terms of (151) are neglected, and positive and negative sequence components take equal value. This results in an impedance matrix of the form

$$\mathbf{Z}_l^{012,\mathrm{s}} \approx \begin{bmatrix} Z_{l,00}^{\mathrm{s}} & 0 & 0 \\ 0 & Z_{l,11}^{\mathrm{s}} & 0 \\ 0 & 0 & Z_{l,11}^{\mathrm{s}} \end{bmatrix}. \tag{152}$$

Therefore, for a line impedance parameterized in terms of only $Z_{l,00}^{\mathrm{s}}$, $Z_{l,11}^{\mathrm{s}}$ (as in (152)) then $\mathbf{Z}_l^{\mathrm{s}}$ has value

$$\mathbf{Z}_l^{\mathrm{s}} = \frac{1}{3}\begin{bmatrix} 2Z_{l,11}^{\mathrm{s}} + Z_{l,00}^{\mathrm{s}} & Z_{l,00}^{\mathrm{s}} - Z_{l,11}^{\mathrm{s}} & Z_{l,00}^{\mathrm{s}} - Z_{l,11}^{\mathrm{s}} \\ Z_{l,00}^{\mathrm{s}} - Z_{l,11}^{\mathrm{s}} & 2Z_{l,11}^{\mathrm{s}} + Z_{l,00}^{\mathrm{s}} & Z_{l,00}^{\mathrm{s}} - Z_{l,11}^{\mathrm{s}} \\ Z_{l,00}^{\mathrm{s}} - Z_{l,11}^{\mathrm{s}} & Z_{l,00}^{\mathrm{s}} - Z_{l,11}^{\mathrm{s}} & 2Z_{l,11}^{\mathrm{s}} + Z_{l,00}^{\mathrm{s}} \end{bmatrix}. \tag{153}$$

A.3 Nomenclature

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- Table of real-valued parameters 20

References

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Symbol	Unit	Vector length	Represents
$\mathbf{U}_i$	(V)	4	Bus i phase-to-ground voltage vector
$\mathbf{U}_i^{\Delta}$	(V)	3	Bus i phase-to-phase voltage vector
$\mathbf{U}_i^{\curlywedge}$	(V)	3	Bus i phase-to-neutral voltage vector
$\mathbf{U}_i^{012}$	(V)	3	Bus i symmetrical component voltage vector
$\mathbf{I}_{lij}$	(A)	4	Line l total current vector in the direction $i \to j$
$\mathbf{I}_{lij}^{\mathrm{s}}$	(A)	4	Line l series current vector in the direction $i \to j$
$\mathbf{I}_{lij}^{\mathrm{sh}}$	(A)	4	Line l shunt current vector in the direction $i \to j$
$\mathbf{S}_{lij}$	(W)	4	Line l total power matrix in the direction $i \to j$
$\mathbf{S}_{lij}^{\mathrm{s}}$	(W)	4	Line l series power matrix in the direction $i \to j$
$\mathbf{S}_d$	(W)	4	Load d power matrix at its bus
$\mathbf{S}_d^{\curlywedge}$	(W)	3	Load d power matrix in wye frame
$\mathbf{S}_d^{\Delta}$	(W)	3	Load d power matrix in delta frame
$\mathbf{I}_d$	(A)	2, 3 or 4	Load d current vector at its bus
$\mathbf{I}_d^{\Delta}$	(A)	3	Load d internal delta current vector (for delta connection)
$\mathbf{S}_g$	(W)	4	Generator g power matrix at its bus
$\mathbf{S}_g^{\curlywedge}$	(W)	3	Generator g power matrix in wye frame
$\mathbf{S}_g^{\Delta}$	(W)	3	Generator g power matrix in delta frame
$\mathbf{I}_g$	(A)	4	Generator g current vector at its bus

Table 18: Complex-valued variables

Symbol	Unit	Size	Represents
$\mathbf{Y}_{lij}^{\mathrm{sh}}$	(S)	$n \times n$	Line l shunt admittance matrix at sending end i
$\mathbf{Y}_{lji}^{\mathrm{sh}}$	(S)	$n \times n$	Line l shunt admittance matrix at receiving end j
$\mathbf{Z}_l^{\mathrm{s}}$	(Ω)	$n \times n$	Line l series impedance matrix
$\mathbf{Y}_h$	(S)	$n \times n$	Bus shunt admittance matrix, to represent neutral grounding, capacitor banks, etc.
$\mathbf{S}_d^{\mathrm{ref},\curlywedge}$	(W+$j$var)	3×1	Load d complex power set-point if wye-connected
$\mathbf{S}_d^{\mathrm{ref},\Delta}$	(W+$j$var)	3×1	Load d complex power set-point if delta-connected
$\mathbf{S}_g^{\mathrm{ref},\curlywedge}$	(W+$j$var)	3×1	Generator g complex power set-point
$\bar{\mathbf{Z}}_x^{\mathrm{s}}$	(Ω/m)	$n \times n$	Per-length impedance
$\bar{\mathbf{Y}}_x^{\mathrm{sh}}$	(S/m)	$n \times n$	Per-length shunt admittance

Table 19: Complex-valued parameters. $n \times n$ matrices will have dimension $1 \leq n \leq 4$ for a four-wire, three phase network.

Symbol	Unit	Size	Represents
$\mathbf{U}_i^{\min}$	(V)	4×1	Bus i phase-to-ground voltage magnitude lower bound
$\mathbf{U}_i^{\max}$	(V)	4×1	Bus i phase-to-ground voltage magnitude upper bound
$\mathbf{U}_i^{\curlywedge,\min}$	(V)	3×1	Bus i phase-to-neutral voltage magnitude lower bound
$\mathbf{U}_i^{\curlywedge,\max}$	(V)	3×1	Bus i phase-to-neutral voltage magnitude upper bound
$\mathbf{U}_i^{012,\min}$	(V)	3×1	Bus i symmetrical component voltage magnitude lower bound
$\mathbf{U}_i^{012,\max}$	(V)	3×1	Bus i symmetrical component voltage magnitude upper bound
$\mathbf{I}_{lij}^{\max}$	(A)	4×1	Line l current magnitude upper bound in the direction $i \rightarrow j$
$\mathbf{S}_{lij}^{\max}$	(VA)	4×1	Line l apparent power upper bound in the direction $i \rightarrow j$
$\mathbf{P}_g^{\min}$	(W)	3×1	Generator g active power lower bound
$\mathbf{P}_g^{\max}$	(W)	3×1	Generator g active power upper bound
$\mathbf{Q}_g^{\min}$	(var)	3×1	Generator g reactive power lower bound
$\mathbf{Q}_g^{\max}$	(var)	3×1	Generator g reactive power upper bound
l_l	(m)	Scalar	Length

Table 20: Real-valued parameters